

Joints of Vertebral Column , its Lesion & Radiology

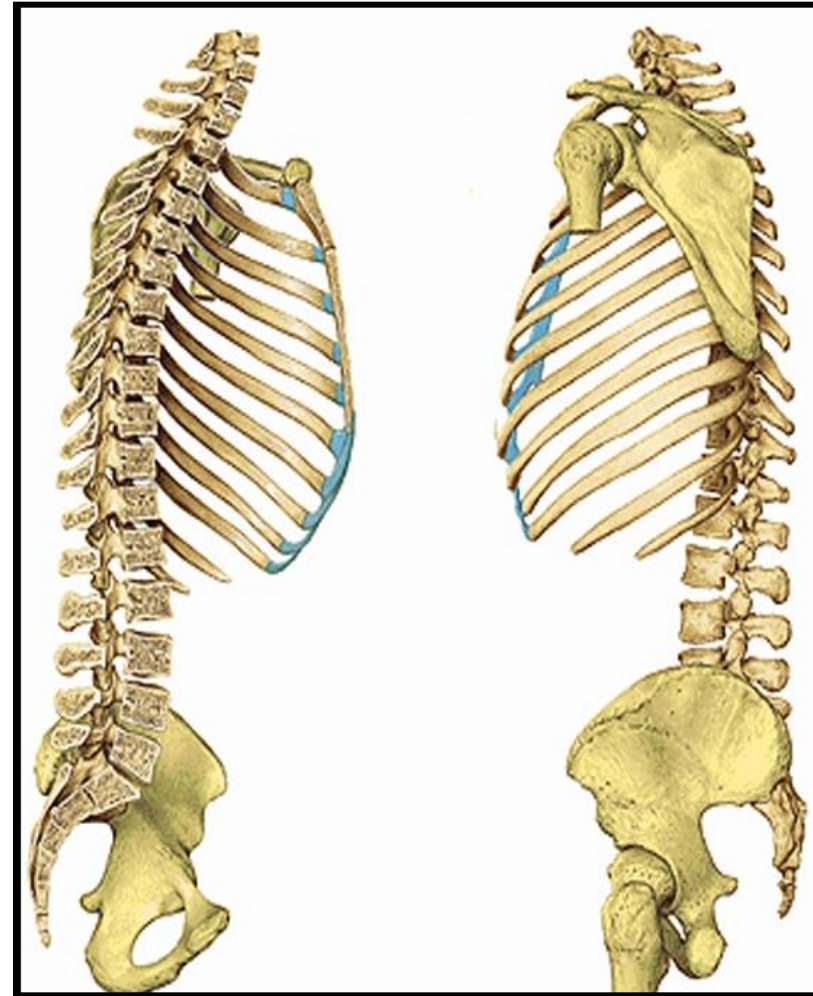


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Objectives

By the end of this lecture, you should be able to:

1. Describe the different intervertebral joints.
2. Recognise the different ligaments of the vertebral column
3. Describe the atlanto-occipital and atlanto-axial joints.
4. Know the ligaments connecting the occipital bone with the axis .
5. Describe the movements and common injuries of the vertebral column.
6. Describe the radiological examination of abnormalities on the vertebral column .
7. Describe the sacroiliac joints regarding (type, articular surface, ligaments and movement)
8. Explain the importance of the sacroiliac joint during standing, movement and sitting.
9. Anatomy of the pelvis (in brief).
10. Knowing the musculoskeletal anatomy on the plain radiographs .
11. Identifying fractures , disc prolapse , abscesses and tumours on radiographs
12. Knowing the basic anatomy on MRI



Joints of the Vertebral column

- ❑ The vertebral column consists of 32-34 vertebrae that articulate with each other by **different types of joints included** :

1. Atlanto-occipital joint

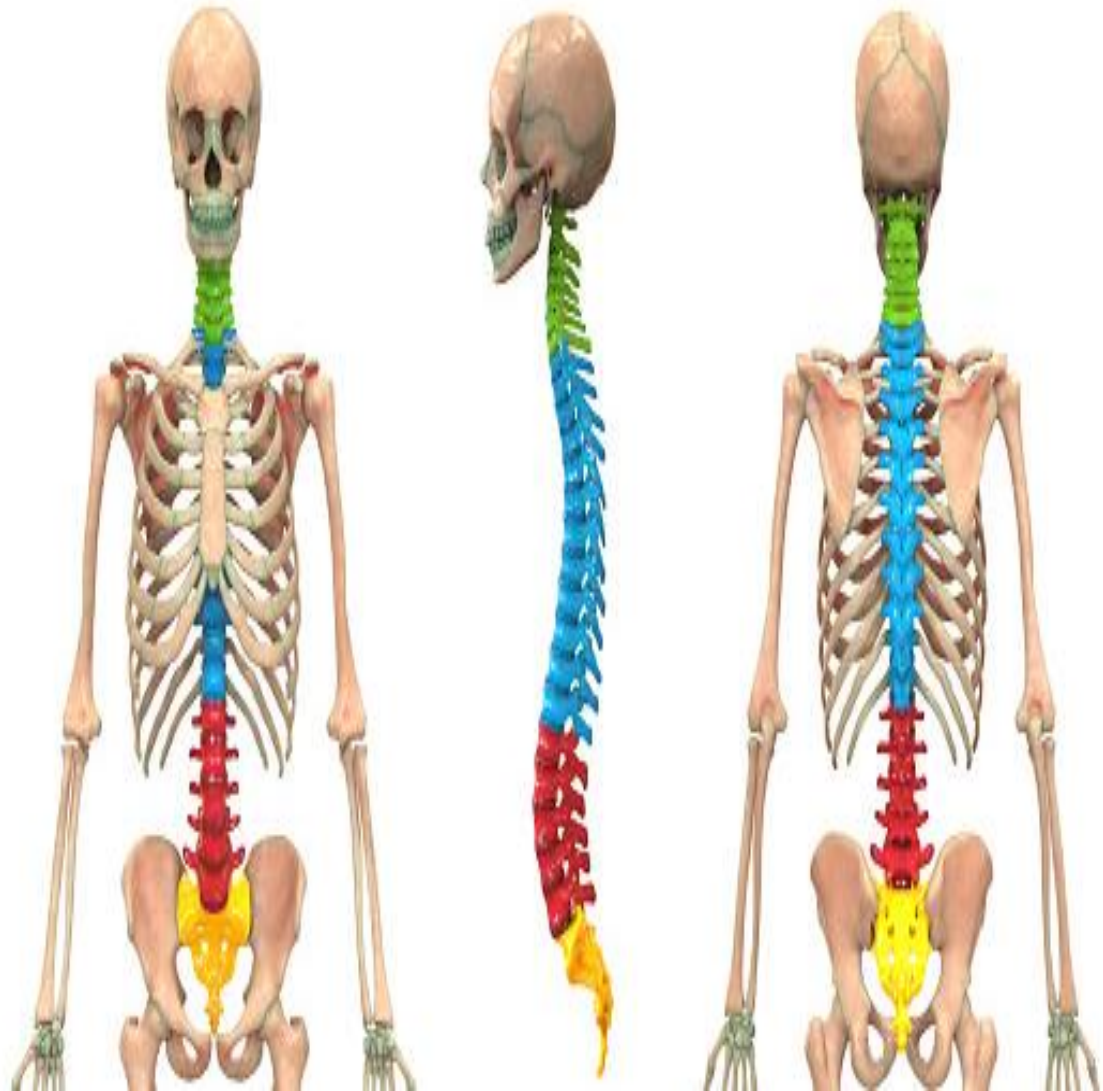
2. Atlanto-axial joint

3. Joints between the other vertebrae below C2 Vertebra

4. Lumbosacral joint

- Sacro-iliac joints (one of types of joints associated with the vertebral column)

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Joints between the other vertebrae below C2 Vertebra

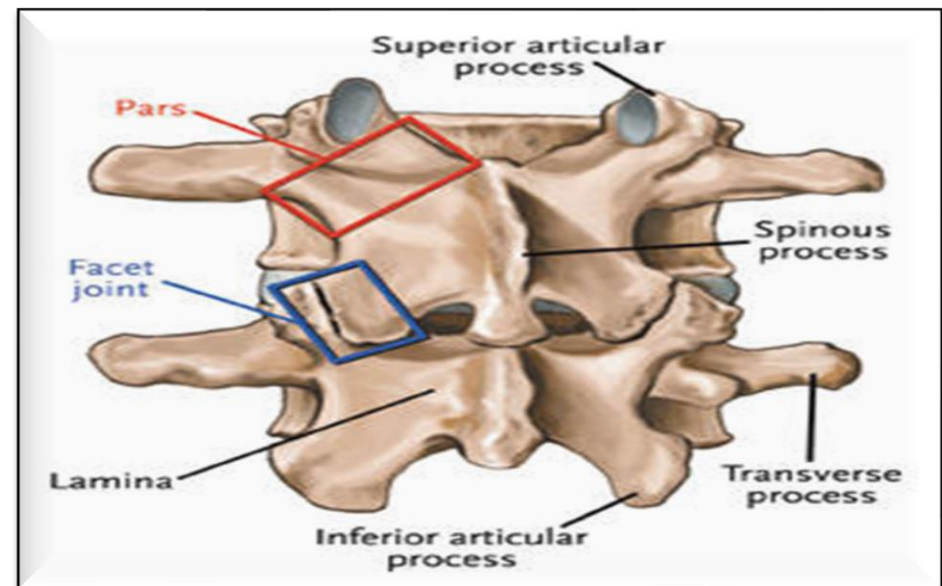
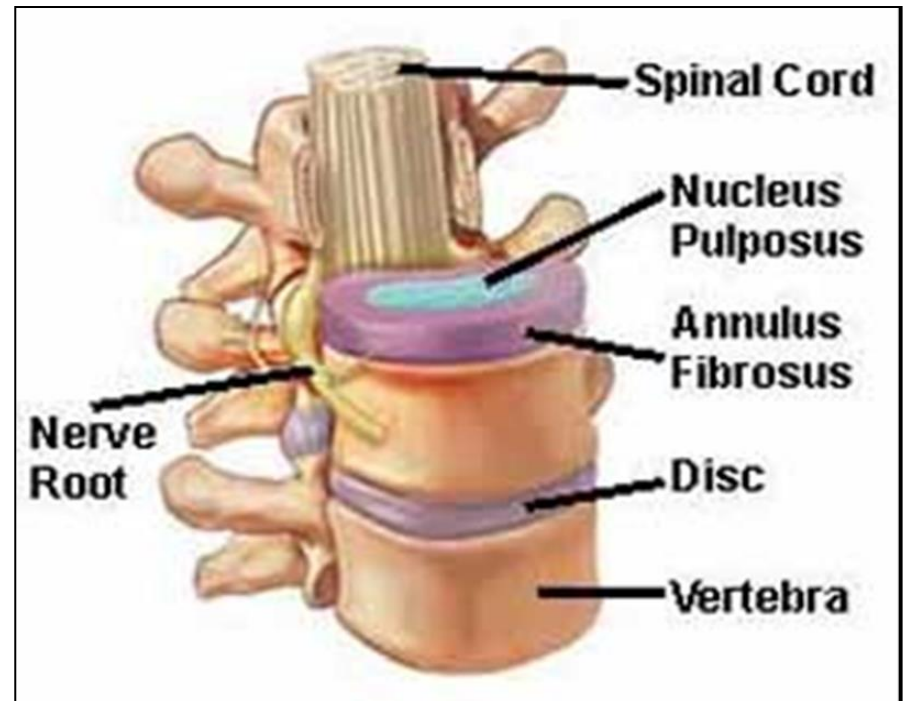
❑ These joints includes:

A. Joint between the vertebral bodies :

- The bodies of the vertebrae articulate with each other by the **secondary cartilaginous joint (Intervertebral Disc)**

B. Joints between the two vertebral arches

- ❑ These are **bilateral plane synovial joints** between the inferior articular processes of the vertebra above and the superior articular processes of the vertebra below
(**Zygapophysial joints**)



Types of Joints of Vertebral Column

1-2ry cartilaginous joints :

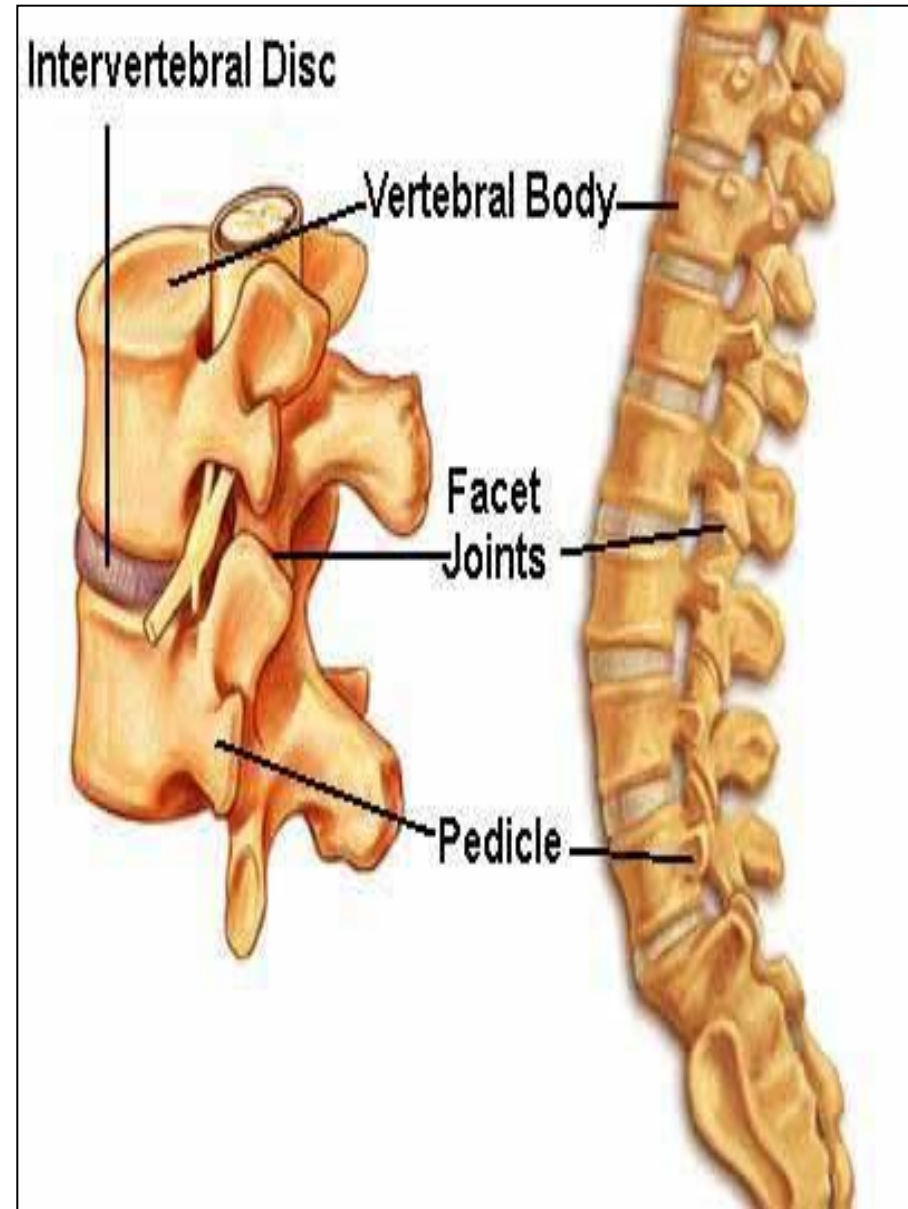
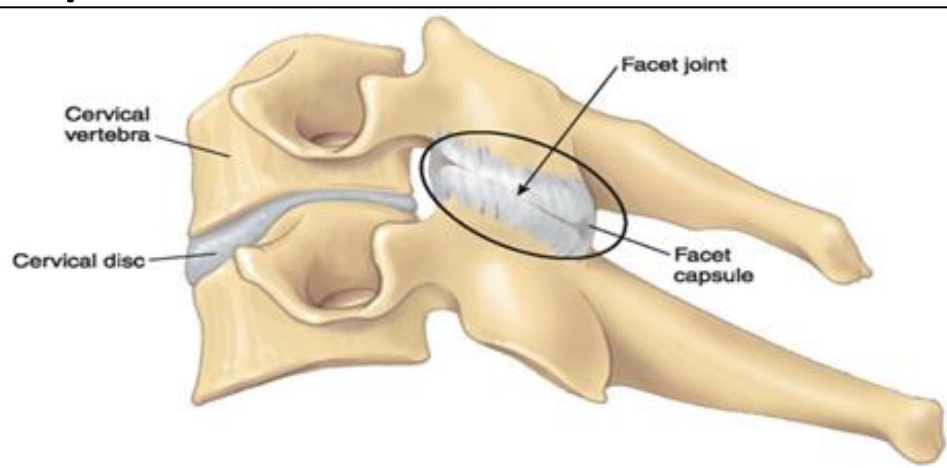
Fibrocartilaginous intervertebral discs between the adjacent vertebral bodies

2- Four Plane synovial joints :

between the articular facets of

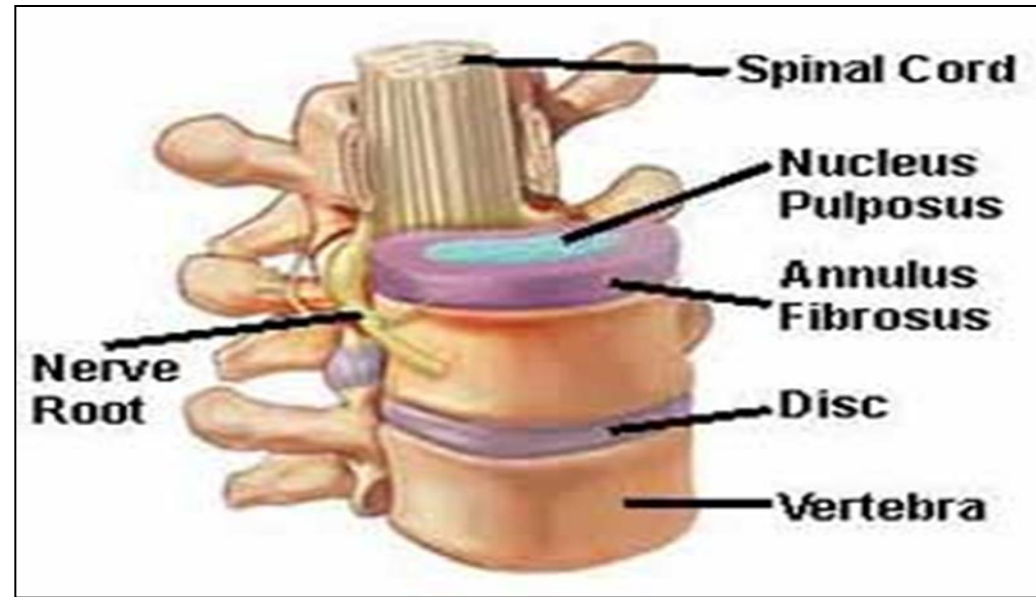
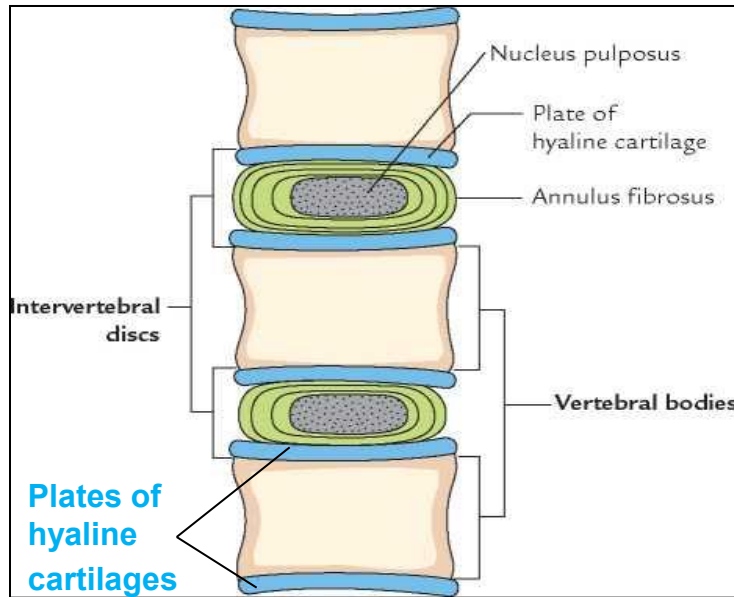
Adjacent vertebrae (2 above & 2 below) .

The articular facets are covered by hyaline cartilage. The joints are surrounded by capsules .



Joints of intervertebral bodies (continued)

I. Intervertebral discs



Intervertebral fibrocartilaginous disc is flat , **oval in shape** , and about a half inch thick . It is formed of 2 components :

1. **Nucleus pulposus** : an **inner soft gelatinous tissue** .(The jelly is partly made of water and gives the disk flexibility and strength) .
2. **Annulus fibrosus** : an **outer flexible fibrous ring** consists of fibrocartilage . It attaches with the **anterior and the posterior longitudinal ligaments**.

II. Plates of hyaline cartilages :

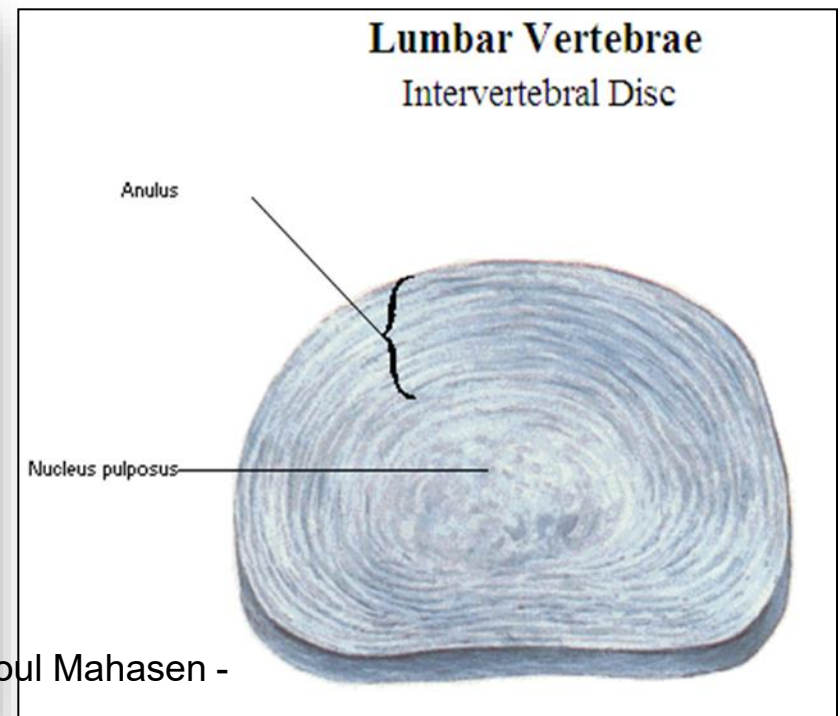
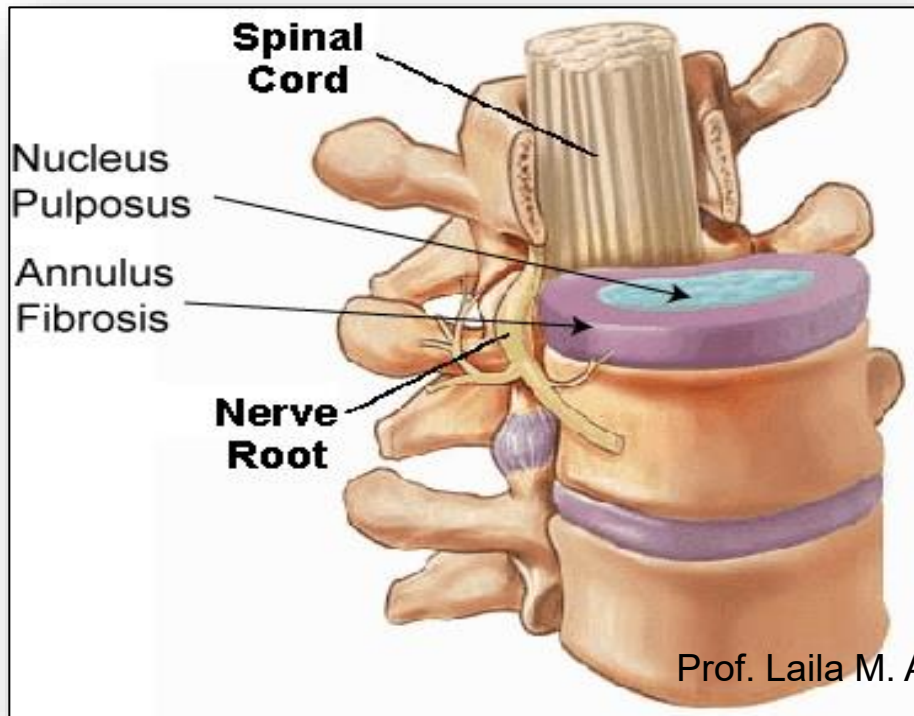
They cover the upper & lower surfaces of vertebral bodies

Functions of the intervertebral disc

*** (The disc is compressible , shock absorber & allows limited movement)**

1. The discs are responsible for $\frac{1}{4}$ of the length of the vertebral column.
2. They provide strong attachments between the vertebral bodies
3. They unite vertebral bodies into a continuous **semirigid** column(shock absorbers)
4. Sudden increase in the compression load on the vertebral column causes flattening the nucleus (allow flexibility of VC)

N.B. Herniation of nucleus pulposus or rupture the annulus fibrosus causing disc prolapse with cord or nerve compression



Ligaments of Vertebral column

1- Anterior longitudinal

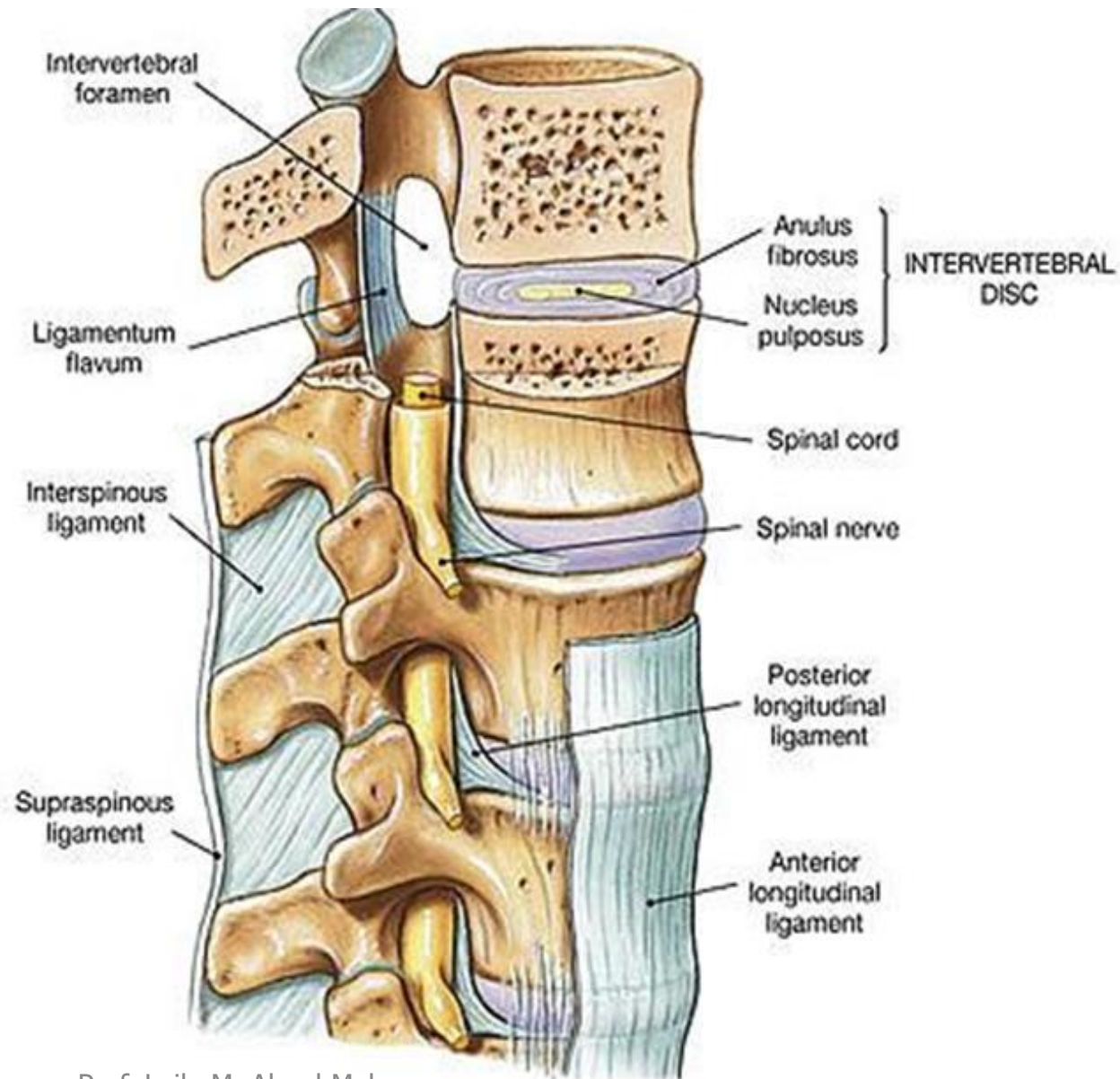
2-Posterior longitudinal

3- Ligamentum Flavum
between 2 adjacent
lamina

4- Intertransverse

5-Interspinous

6- supraspinous



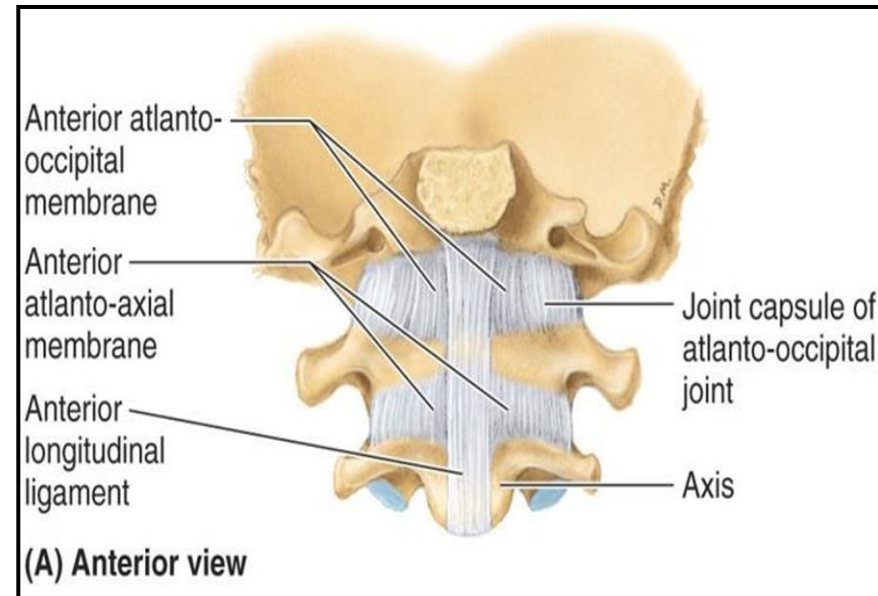
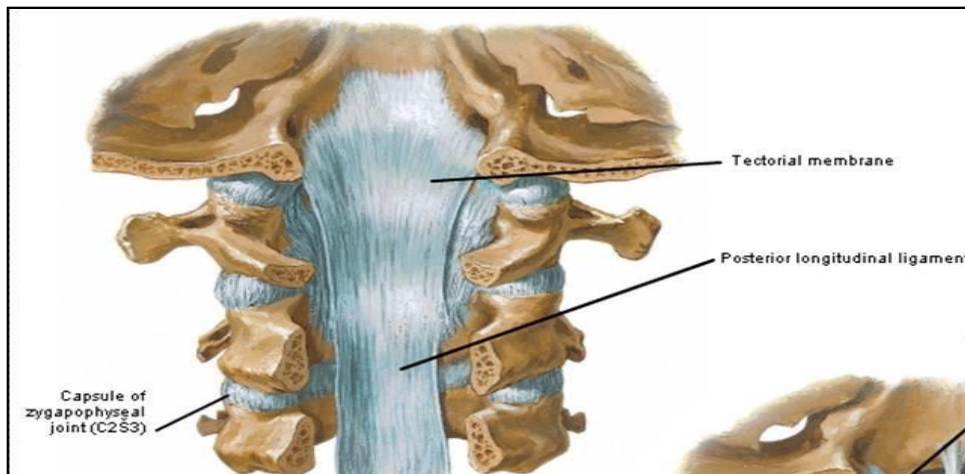
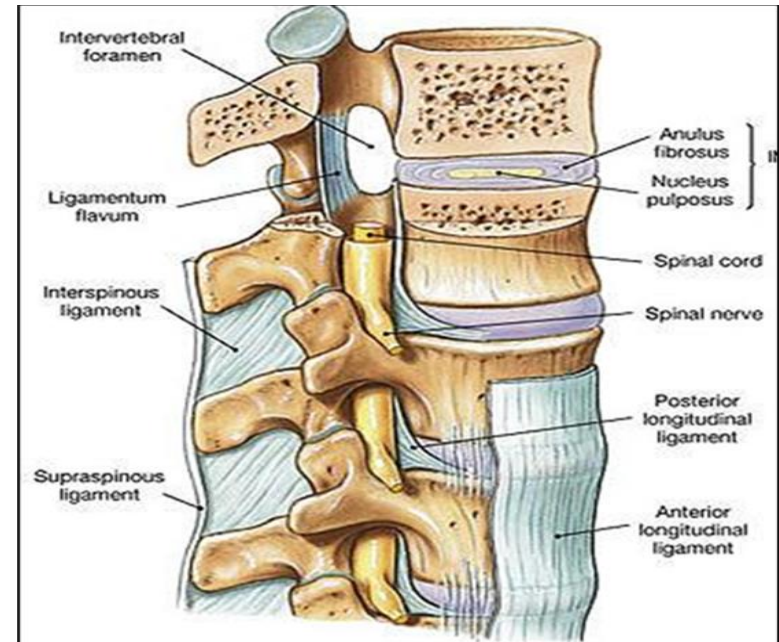
Ligaments of Vertebral column

1. Anterior longitudinal ligament :

- It is strongly attaches to the **front and sides** of the vertebral bodies and the intervertebral discs.
- Its is continuous upward as the anterior atlanto-occipital membrane.

2. Posterior longitudinal ligament :

- It is a weak ligament. It is situated in the vertebral canal and attached to the **posterior surfaces of the vertebral bodies and the intervertebral discs**.
- Its is continuous upward as the membrana tectoria opposite the **upper two cervical vertebrae**.



Ligaments of Vertebral column(continued)

3. Supraspinous ligament:

It connects between the **tips of the adjacent spines**.

4. Interspinous ligament:

It connects the **length of the adjacent spines**.

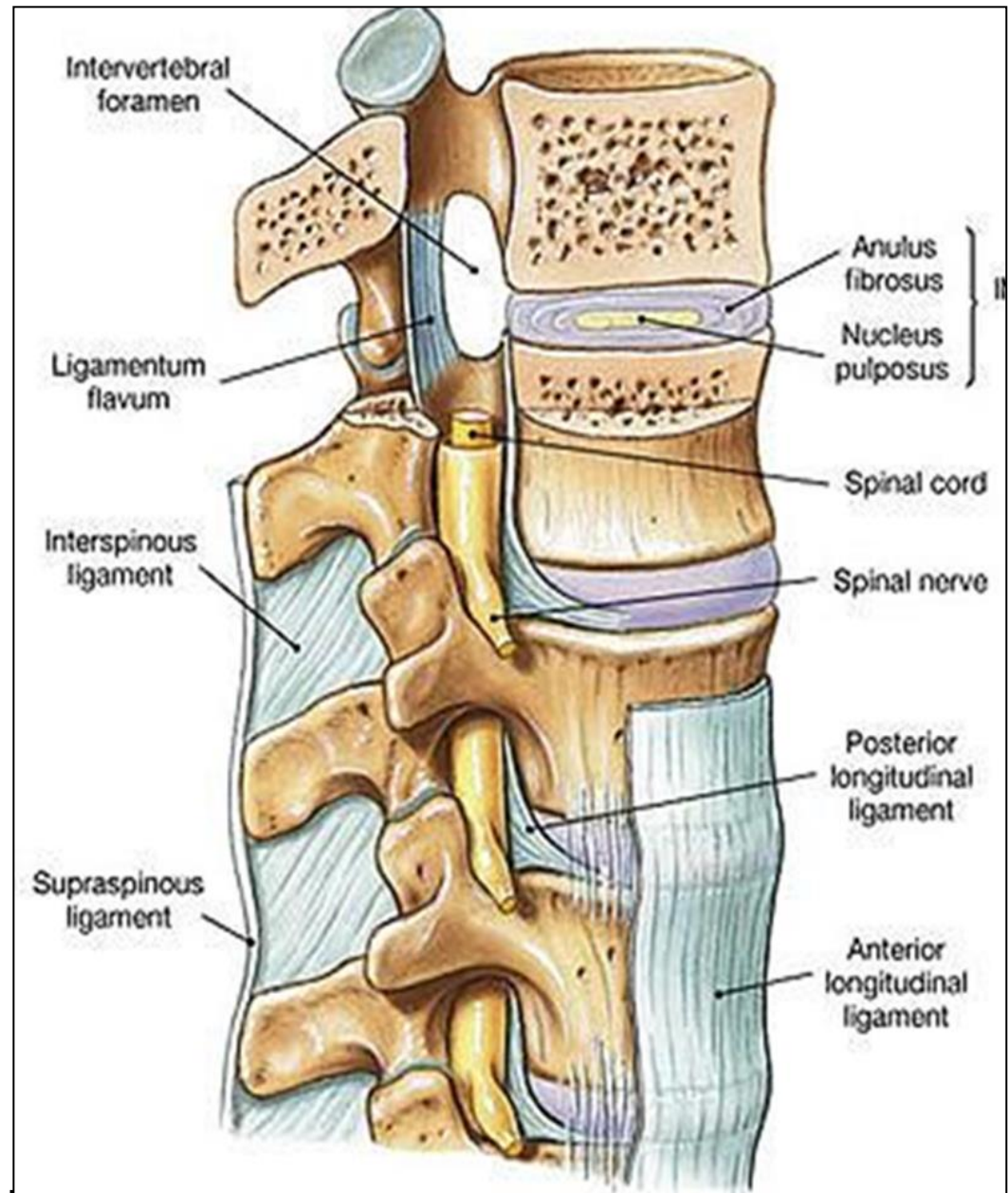
➤ **The supraspinous and the interspinous ligaments :** are greatly thickened in the cervical region to form the **ligamentum nuchae**.

5. Intertransverse ligaments:

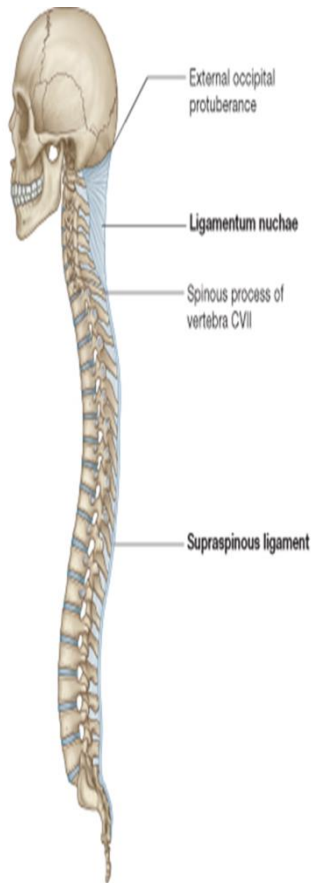
They connect the **transverse processes** .

6. Ligamentum flavum:

It connects the **laminae of the adjacent vertebrae**.



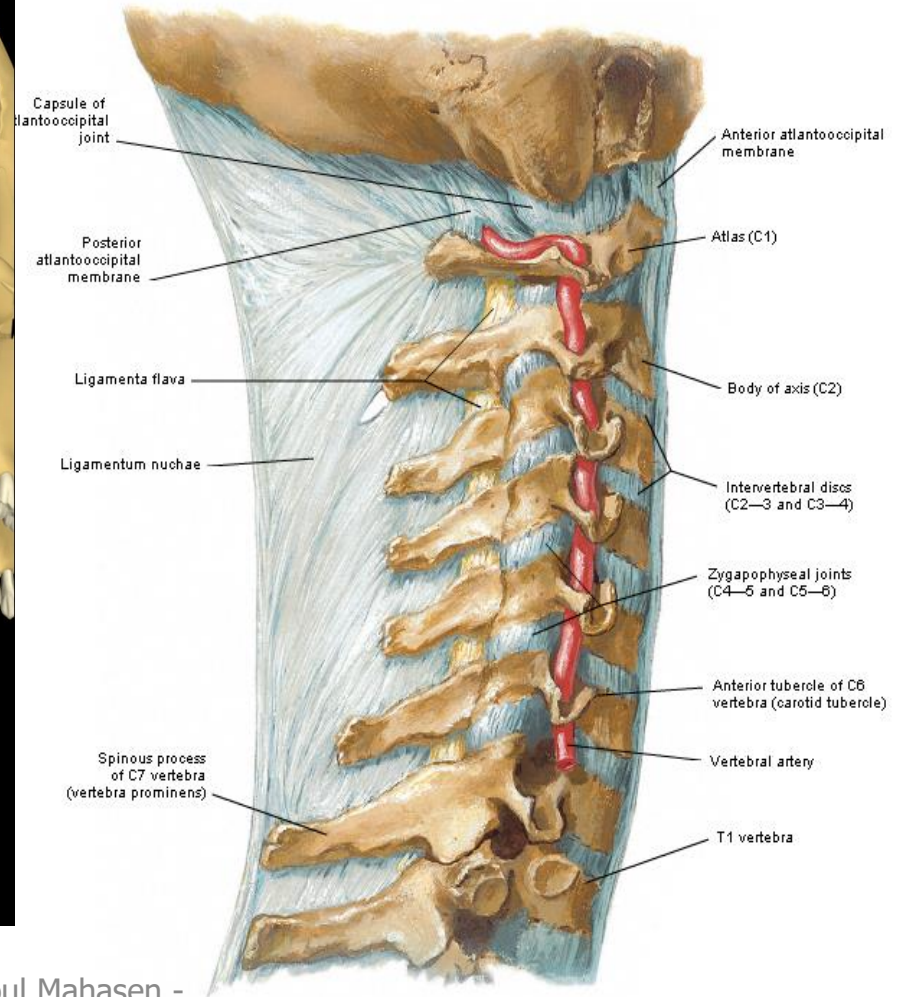
Nuchal ligament (extends between external occipital protuberance to spine of C6)



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External Craniocervical Ligaments
Right Lateral View



Atlanto-occipital joints

Type : an ellipsoid synovial joint between occipital condyle & the superior facets of lateral mass of atlas on each side.

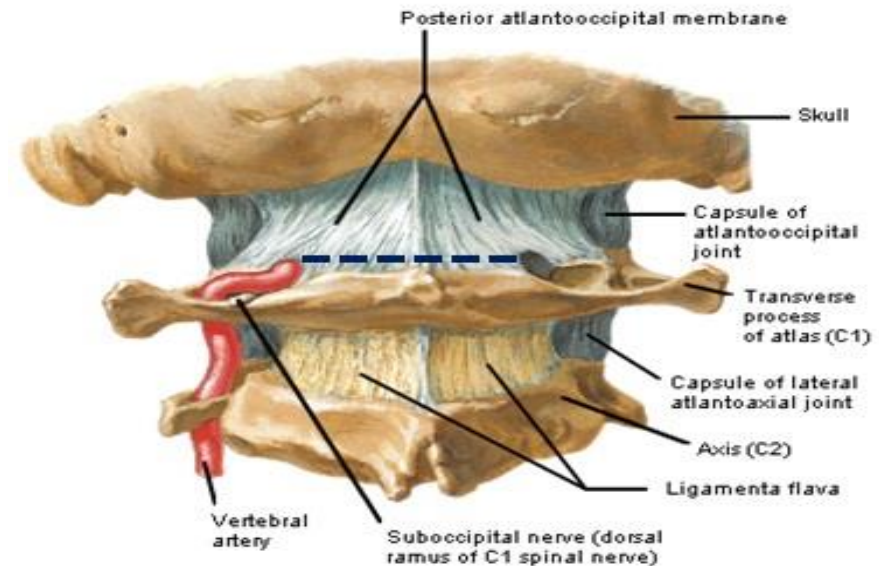
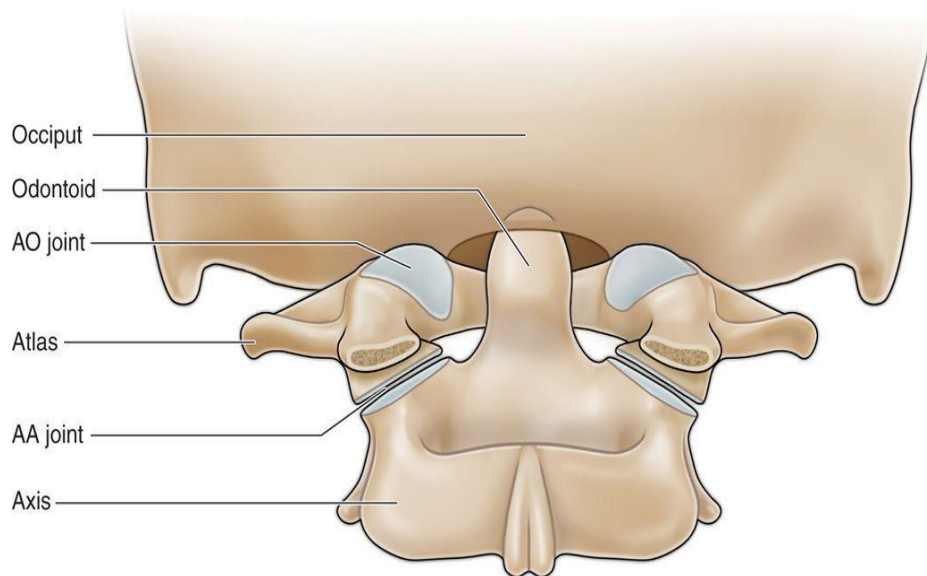
Ligaments (membranes) : 3 major

1. Anterior atlanto-occipital membrane connects anterior arch of atlas with anterior margin of foramen magnum. It is the continuation of the ant. Long. ligament.

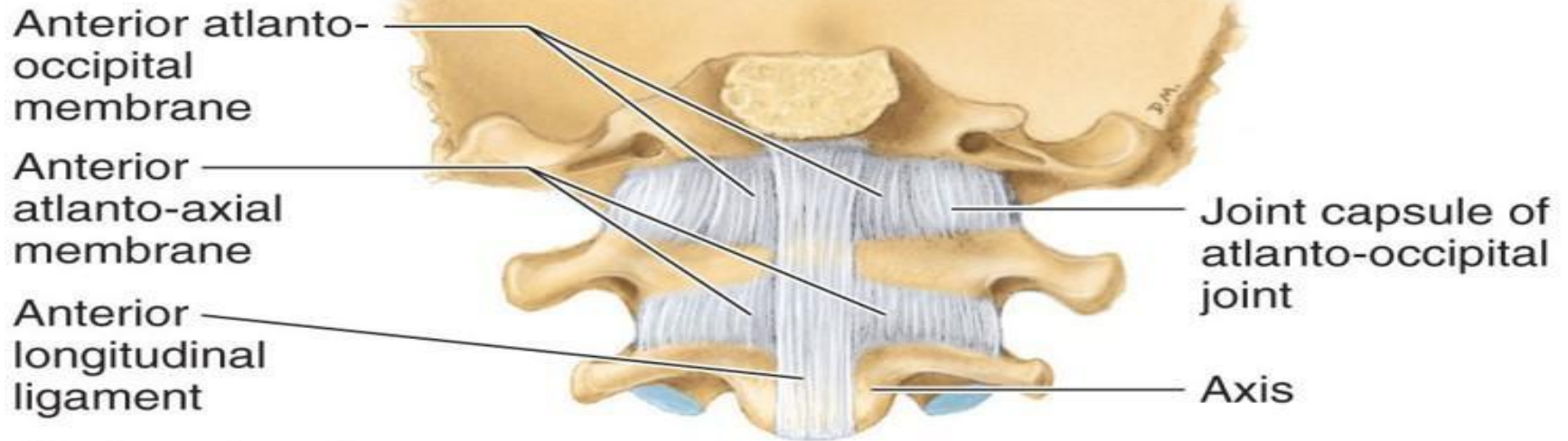
2. Posterior atlanto-occipital membrane connects posterior arch of atlas with posterior margin of foramen magnum

3. Tectorial membrane : is the upper part of posterior longitudinal membrane .

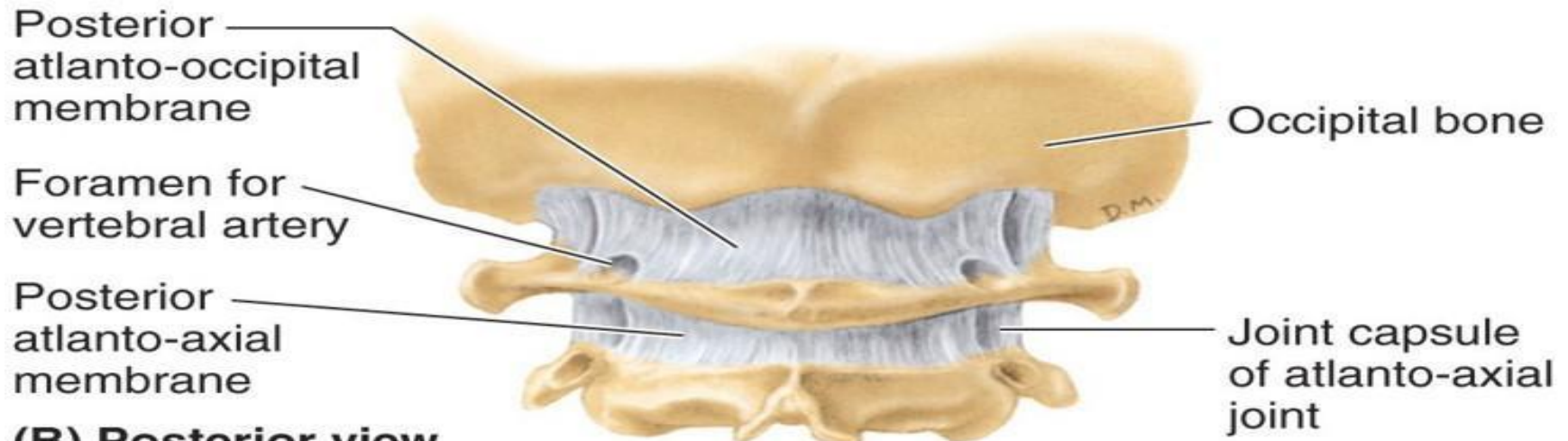
Movements : Nodding movement (YES) : Flexion , Extension & Lateral Flexion
No rotatory movement .



Ligaments of atlanto-occipital joint & atlanto-axial joint



(A) Anterior view



(B) Posterior view

Atlanto-axial joints

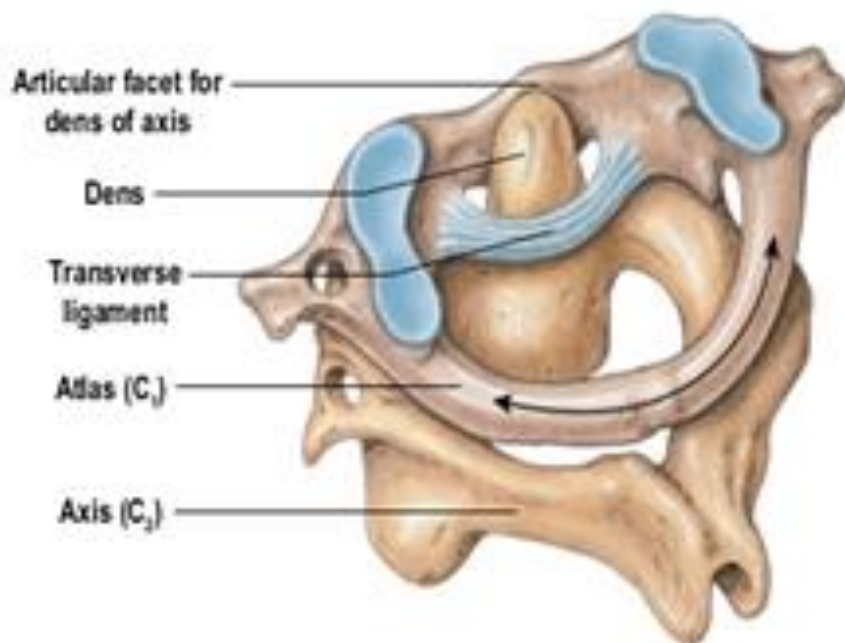
1. Two Lateral atlanto-axial joints : two plane synovial joints between the inferior facets of lateral masses of atlas & the superior facets of the axis .

2. Median atlanto-axial joint : a pivot synovial joint between dens of axis & a ring formed by anterior arch & transverse ligament of atlas.

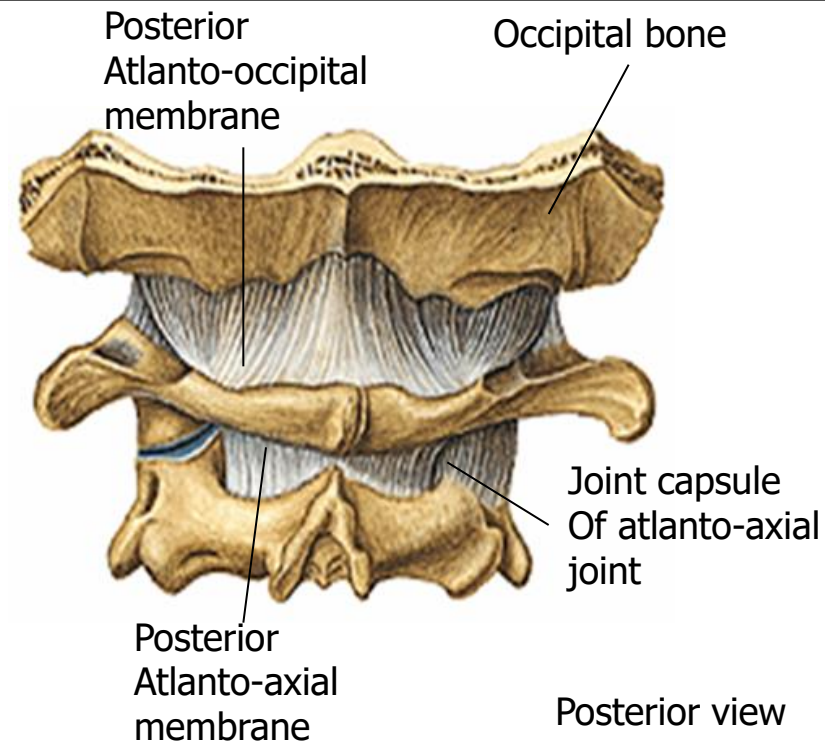
* **Transverse ligament of atlas** holds the dens in its place .

Movements : they permit side to side **(rotatory) movements(NO)**

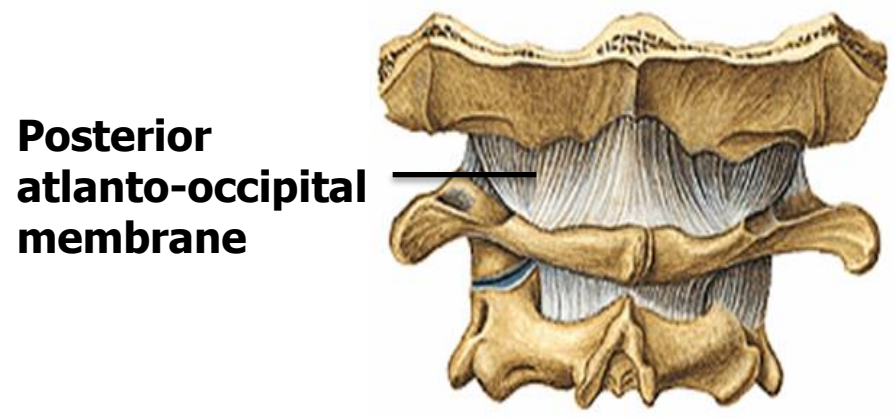
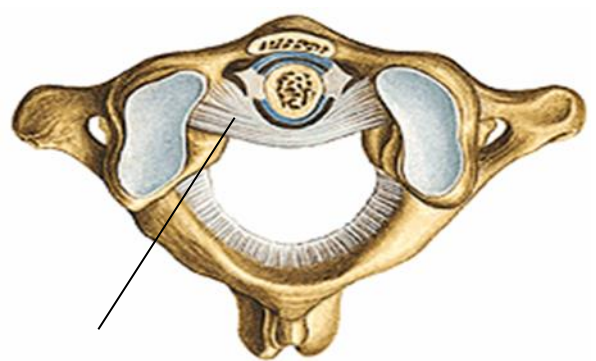
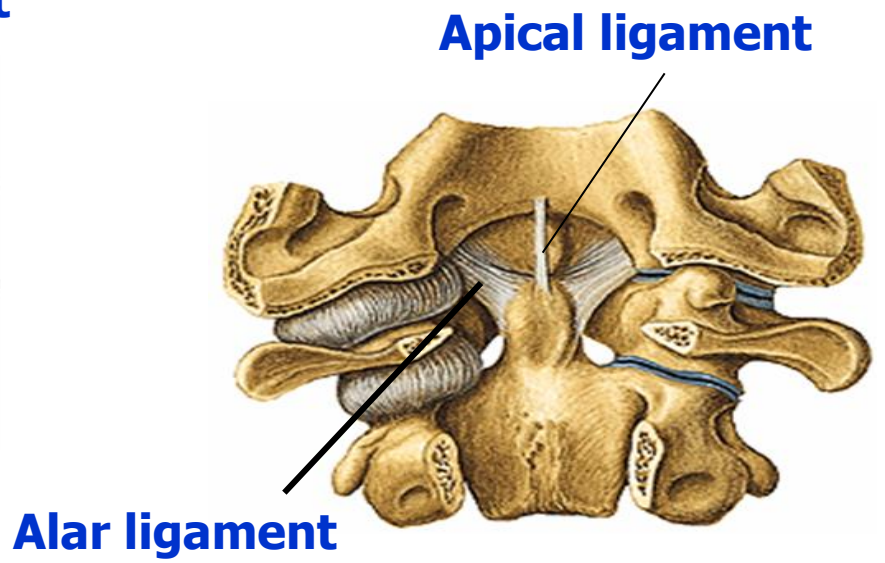
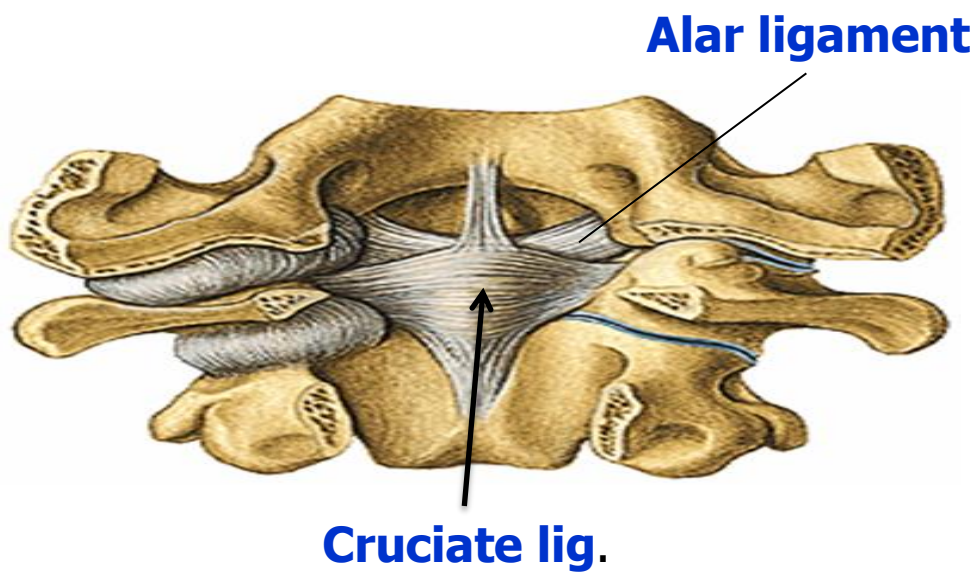
****This joint is dislocated during the hanging process & the odontoid process press on the medulla oblongata which contains the vital centres leads to death .**



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Ligaments of Atlanto-occipital Joint & Atlanto-axial Joint

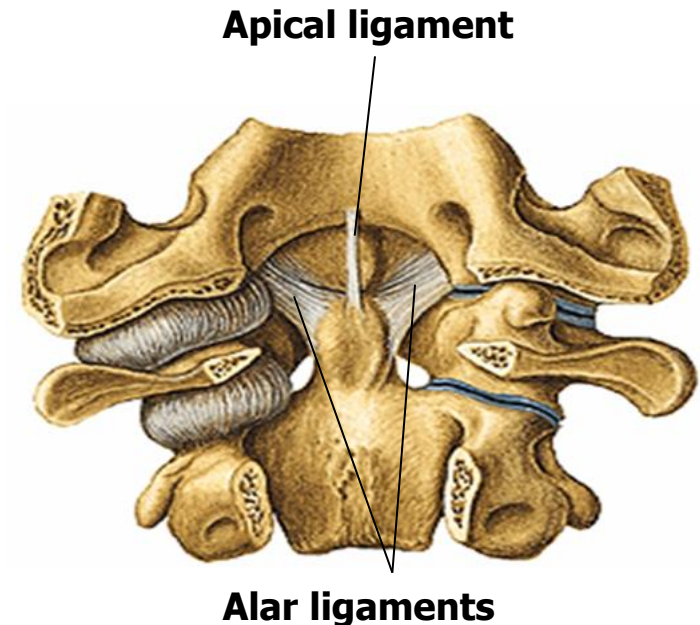
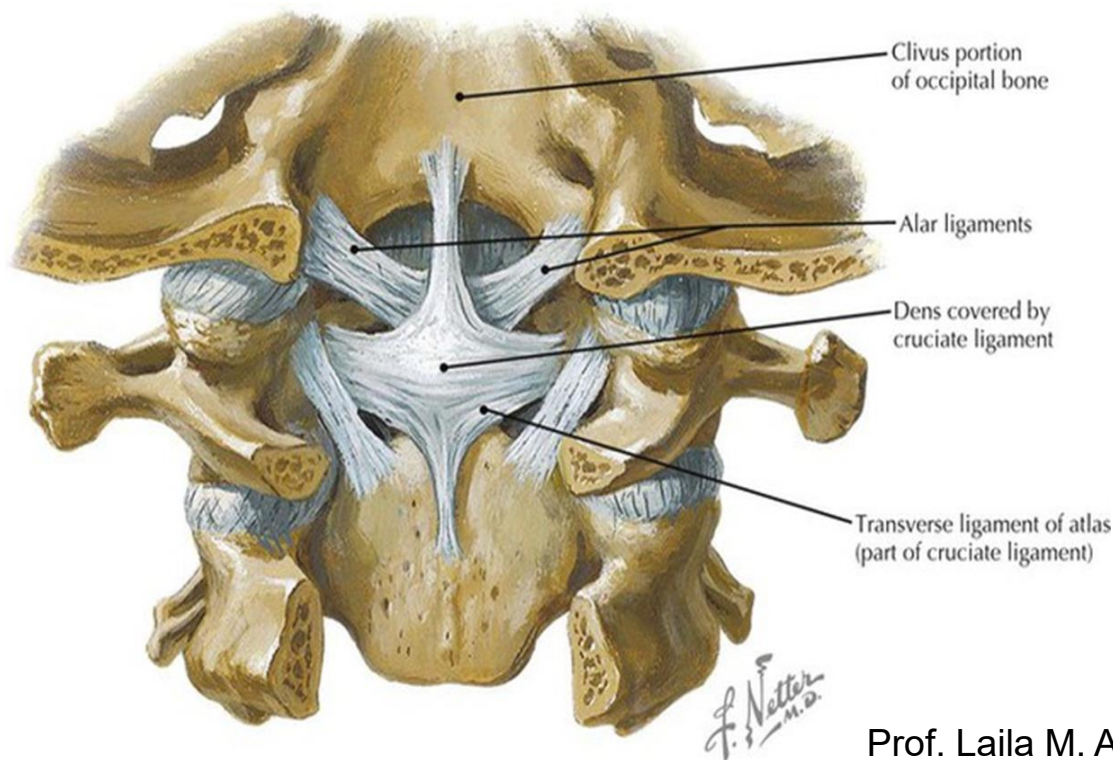


Ligaments between axis & occipital bone

(Apical ligament -alar ligament & cruciate ligament)

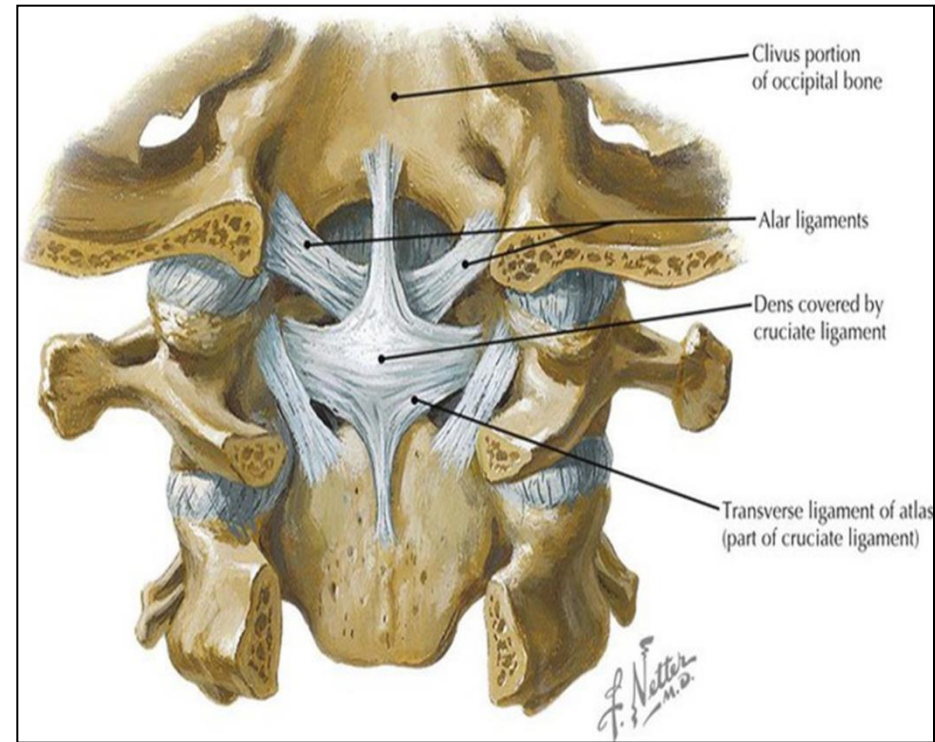
1- Apical ligament: it connects between the apex of odontoid process of axis and the anterior margin of foramen magnum

2- Alar Ligaments: it connects the side of the odontoid process and the medial side of the occipital condyle (lateral margin of foramen magnum) .



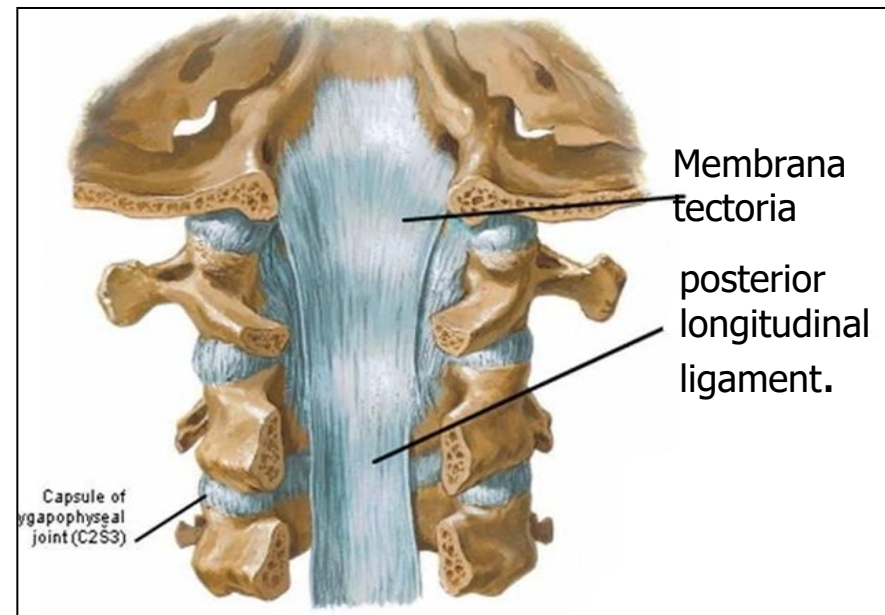
3- Cruciate ligament: it consists of vertical and transverse parts.

- ❑ **The transverse part** connects between the inner aspect of the lateral masses of the atlas and support the odontoid process against the anterior arch of atlas.
- ❑ **The vertical part** connects between the posterior surface of the body of the axis and the anterior margin of the foramen magnum.



4- Membrana tectoria:

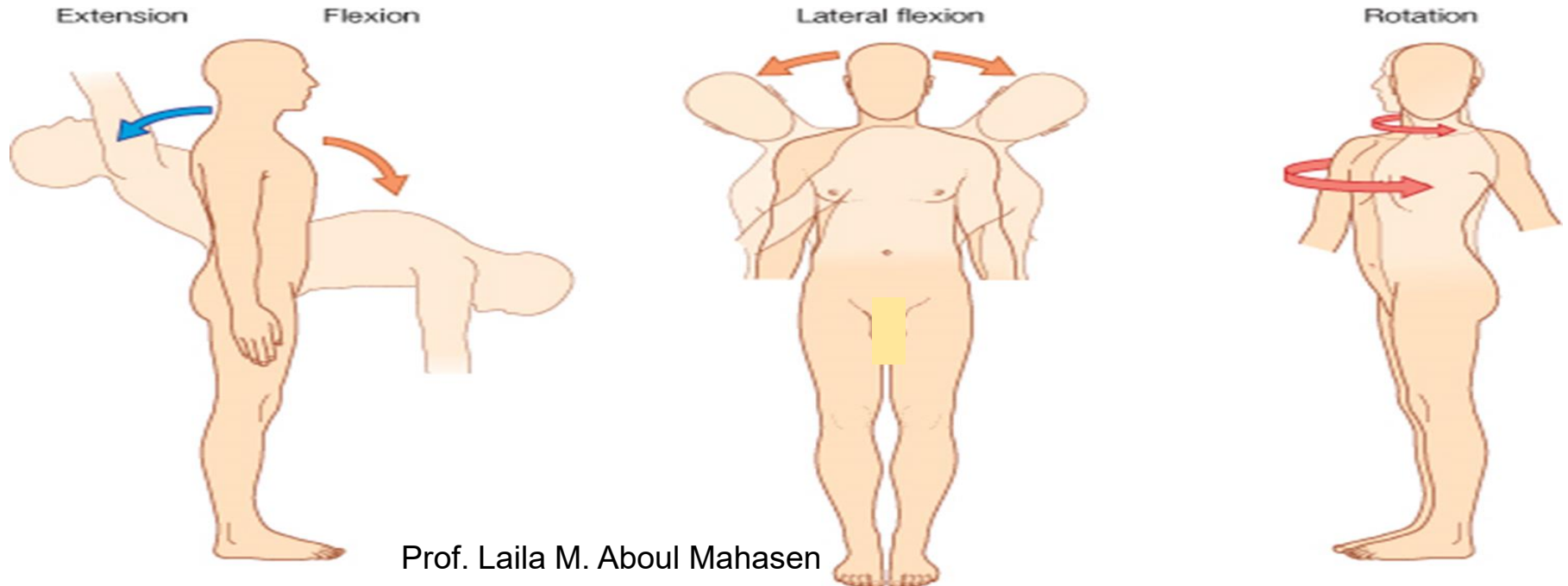
- ❑ It connects between the posterior surface of the axis and the occipital bone.
- ❑ It is the continuation of the posterior longitudinal ligament.
- ❑ It covers the back of the odontoid process, alar and apical ligaments.
- ❑ **Movements: Rotatory movements of the head**



Movements of the vertebral Column

The vertebral column has a remarkable degree of mobility as follows:

- ❑ **1- Flexion:** by longus Cervicis, scalenus anterior, sternomastoid, Rectus abdominis and Psoas major muscles
 - ❑ **2- Extension:** by postvertebral muscles
 - ❑ **3- Lateral flexion:** by scalenus anterior, scalenus medius, Trapezius, sternomastoid, quadratus Lumborum & anterolateral abdominal muscles
 - ❑ **4- Rotation:** by sternomastoid, Splenius, rotators, Semispinalis.
 - ❑ **5- Circumduction:** is the combination of all these movements.
- ❖ **The movements are extensive in the cervical (most) and lumbar regions and restricted in thoracic region.**



Lumbosacral joint

Type : is a 2ry cartilaginous joint .

The intervening cartilaginous disc : is very thick allowing **more movement** in this joint as compared with other intervertebral joints .

Ligaments : has 2 **special** ligaments in addition to the **typical ligaments** of the intervertebral joints .

The typical intervertebral ligaments :

1- **Anterior & posterior longitudinal ligaments** : uniting the vertebral bodies

2- **Capsular ligaments** : connect the articular processes

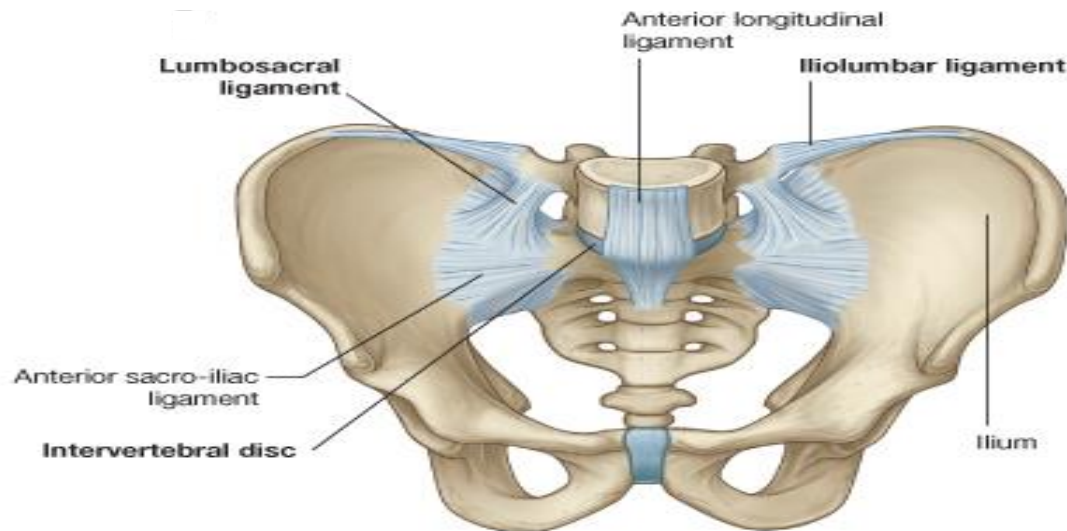
3- **Supraspinous & interspinous ligaments** : uniting the vertebral spines

4- **Ligamenta flava** : uniting the vertebral laminae

The special ligaments : **Iliolumbar & lateral lumbosacral**

1- **Iliolumbar ligament** : a strong ligament connects the transverse process of the L5 with the inner lip of iliac crest . It supports also upper part of sacroiliac joint .

2- **Lateral lumbosacral ligament** : connects the transverse process of the L5 with the ala of the sacrum .



The divisions of the pelvis

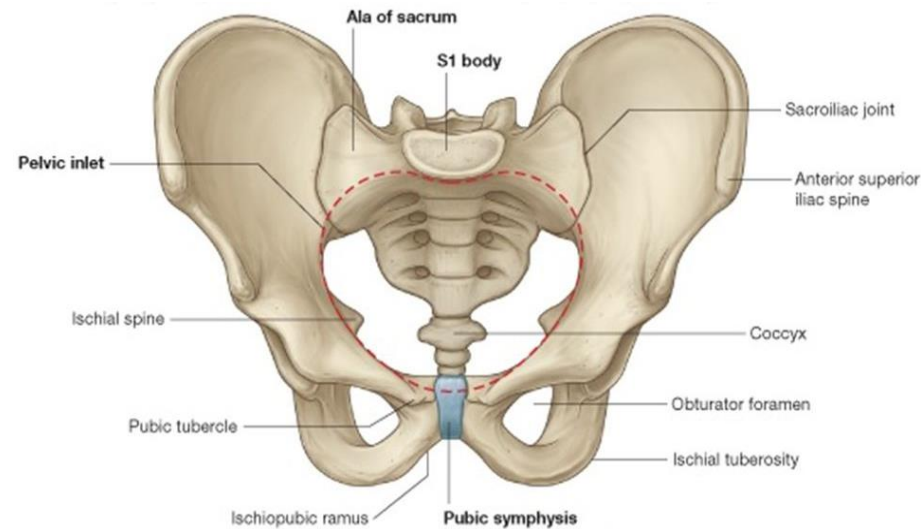
It is divided into 2 parts by the **plane of the pelvic brim (pelvic inlet)** .

- Above the brim is the **false pelvis, or greater pelvis**
- Below the brim is the **true pelvis, or lesser pelvis**.

Boundaries of pelvic brim :

It is formed by :

1. The sacral promontory **posteriorly** .
2. The ilio-pectineal lines **laterally** .
3. The symphysis pubis **anteriorly** .



(c) Anterior view of false pelvis (pink)

It is the expanded part of the pelvis .
It is bounded laterally by the 2 iliac fossae .



(d) Anterior view of true pelvis (blue)

It is a bowl-shaped structure contains the pelvic viscera . It has inlet , outlet & cavity

The True Pelvis (lesser pelvis)

Definition : it is a bowl-shaped structure that contains & protects the lower parts of the intestinal and urinary tracts and the internal organs of reproduction.

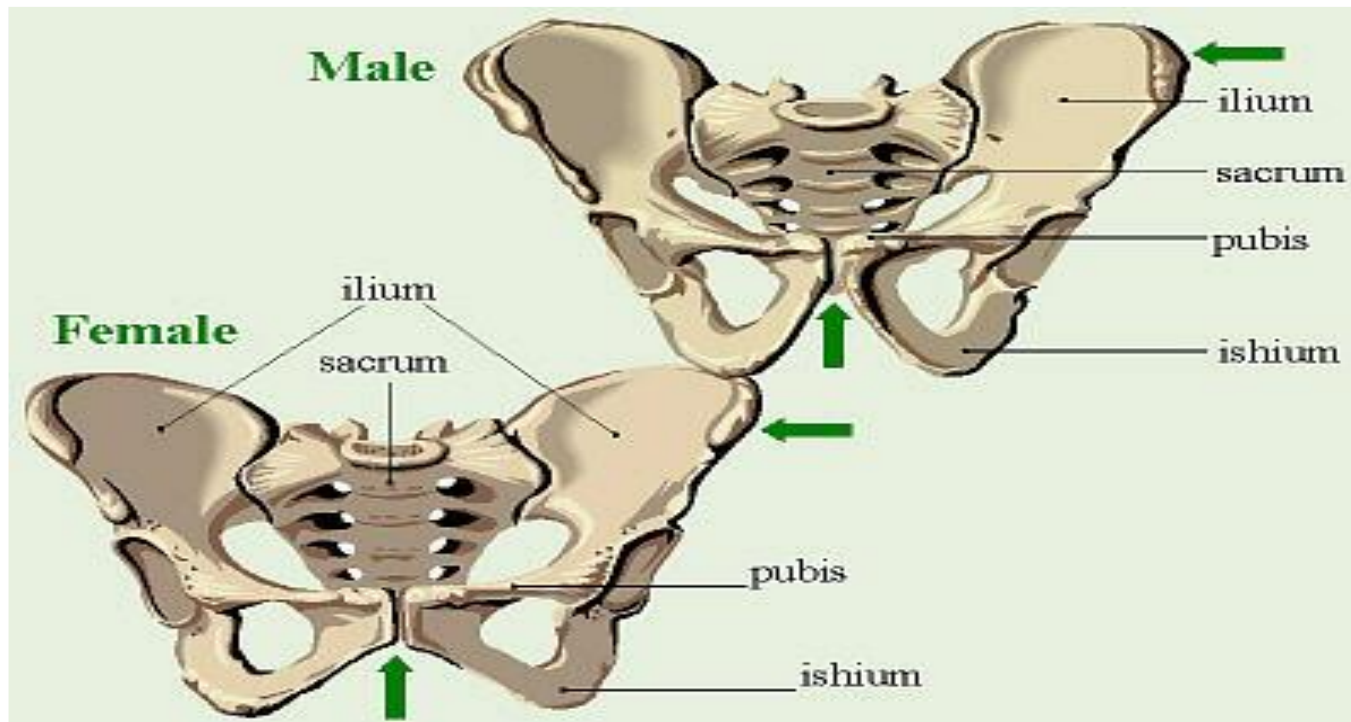
Parts : it has an **inlet**, an **outlet** and a **cavity**.

1-The **pelvic inlet** is the pelvic brim.

2-The **pelvic outlet** is bounded by :

- a. **Posteriorly** : the coccyx
- b. **Anteriorly** : the pubic arch.
- c. **Laterally** : the ischial tuberosities , sacrotuberous and sacrospinous ligaments .

3-The **pelvic cavity** lies between the inlet and the outlet. It is a short curved canal .



Right & Left Sacroiliac joints

They are **very strong synovial joints** between the auricular surfaces of the sacrum and the iliac bones .

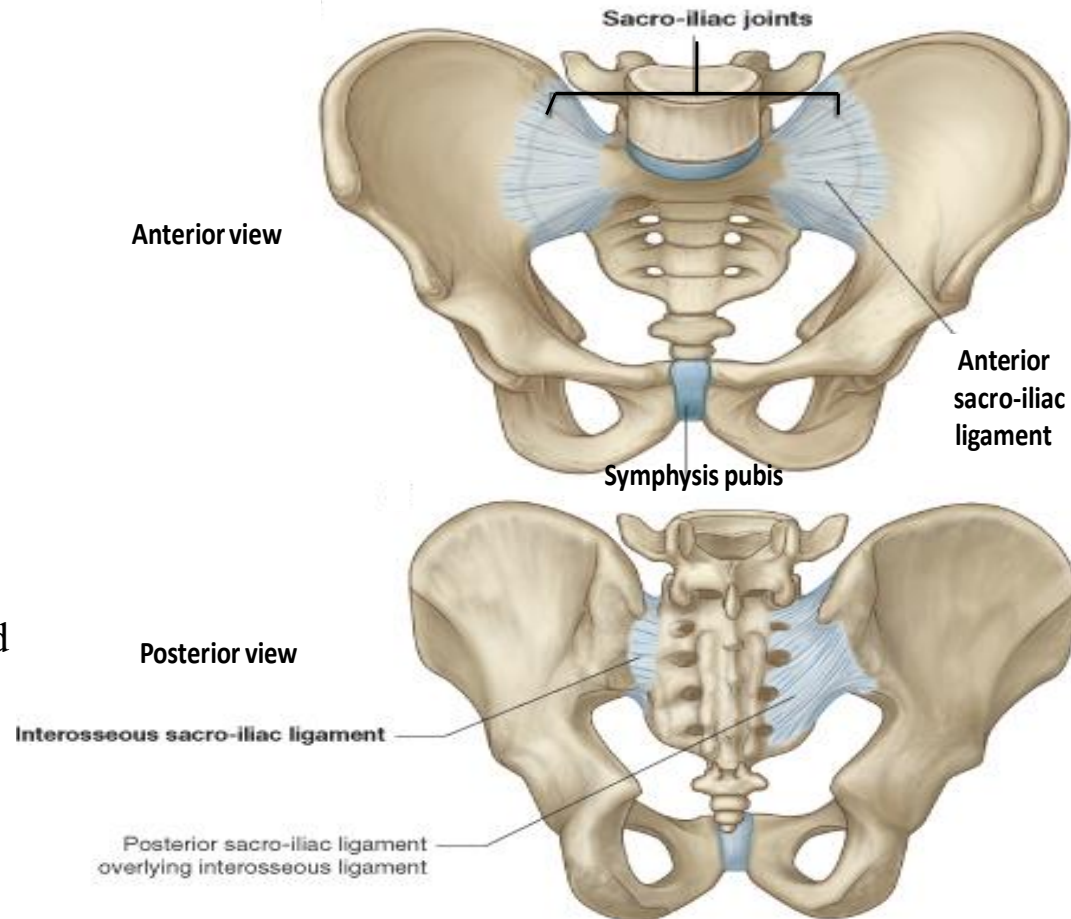
Type : it is a synovial joint of an irregular elevations and depressions that produce **interlocking** of the two bones .

Ligaments : There are 3 ligaments (posterior , interosseous & anterior) :

- 1- **The very strong interosseous sacroiliac ligament** : connects the rough area behind the auricular surface of the sacrum with the iliac tuberosities
- 2- **The strong posterior sacroiliac ligament** connects the posterior superior iliac spine with the transverse tubercles of posterior surface of the sacrum .
- 3- The **anterior sacroiliac ligament** is thin and is placed in front of the joint.

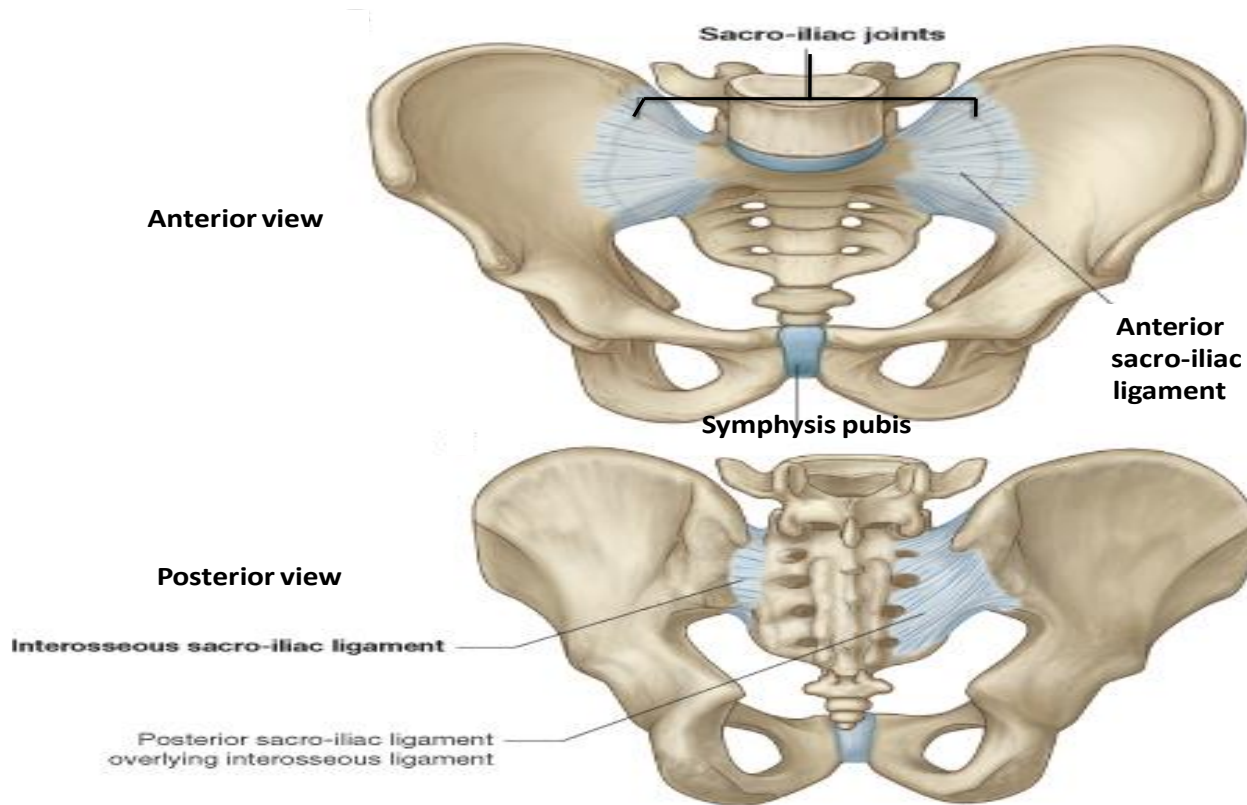
Accessory ligaments :

1. Iliolumbar ligament
2. Sacrotuberous ligament
3. Sacrospinous ligament



Movements of Right & Left Sacroiliac Joints

- A small but limited amount of movement is possible at these joints. The ligaments relaxed gradually during pregnancy to **widen the pelvis during labour**.
- **The main function** is to transmit the weight of the body from the vertebral column to the bony pelvis.



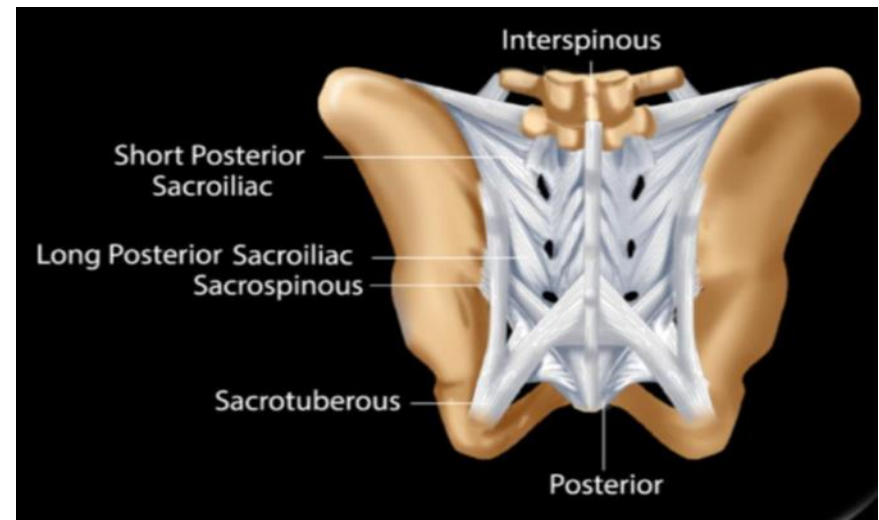
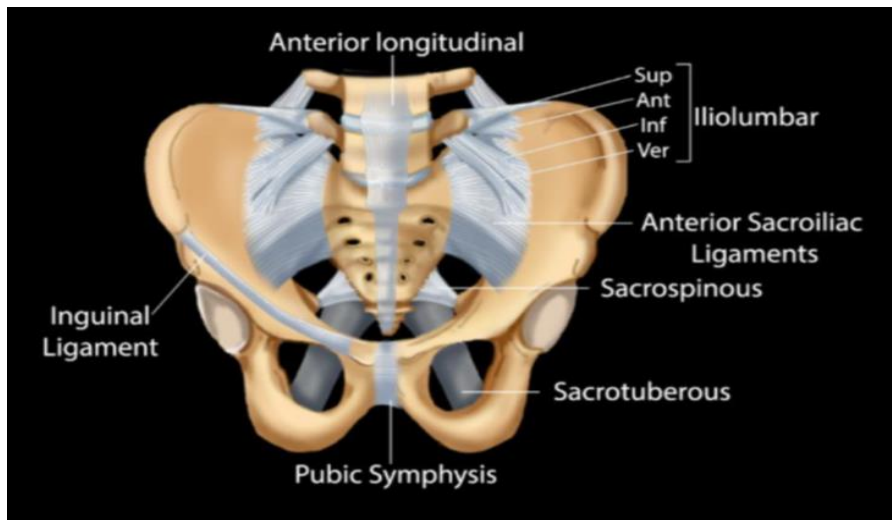
Stability of sacroiliac joints

Superior support: Iliolumbar ligament of lumbosacral joint .

Anterior support: Anterior sacroiliac ligament .

Posterior support: Posterior sacroiliac & Interosseous ligaments

Accessory ligaments: Sacrotuberous & Sacrospinous ligaments



Sacrotuberous & Sacrospinous ligaments

- Help stabilize the sacrum on the pelvic bone by preventing from forward tilting of the sacral promontory.
- They also converting greater and lesser sciatic notch into greater and lesser sciatic foramina.

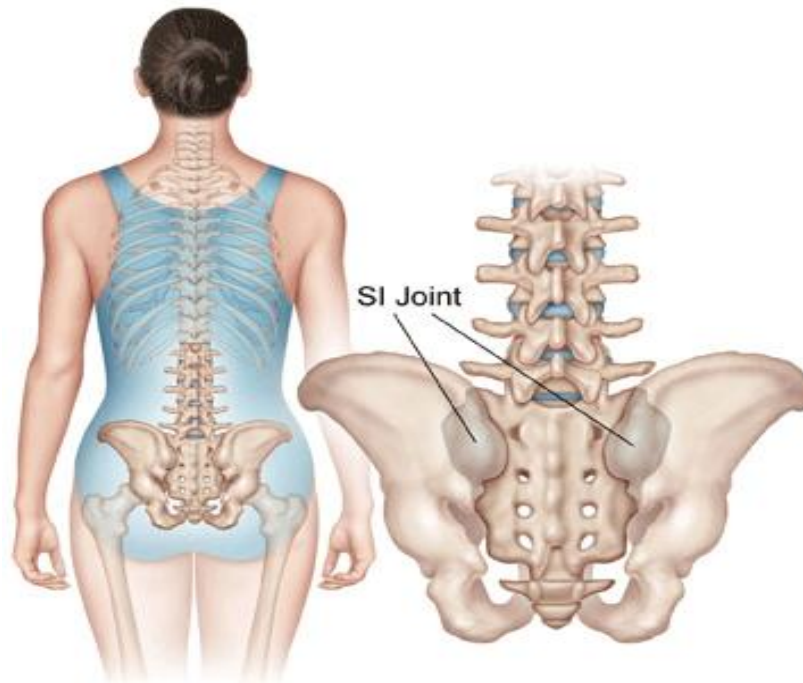
Functions of sacroiliac joints

- The sacroiliac joints are essential for effective load transfer between the spine and the lower limbs .
- It functions both as a shock absorber for the spine above and converts torque from the lower limbs into the rest of the body. So , it provides stability and attenuate forces to the lower limbs.

Why the SI joints are so important ?

Every time you walk, your SI joints distribute the shock of motion across the pelvis, thereby reducing strain on the spine. The joints also help stabilize your body, enabling you to maintain an upright posture while walking or running.

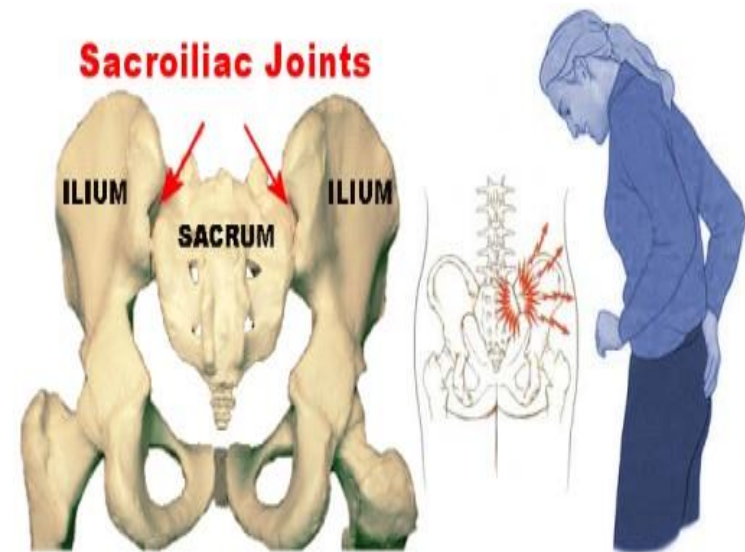
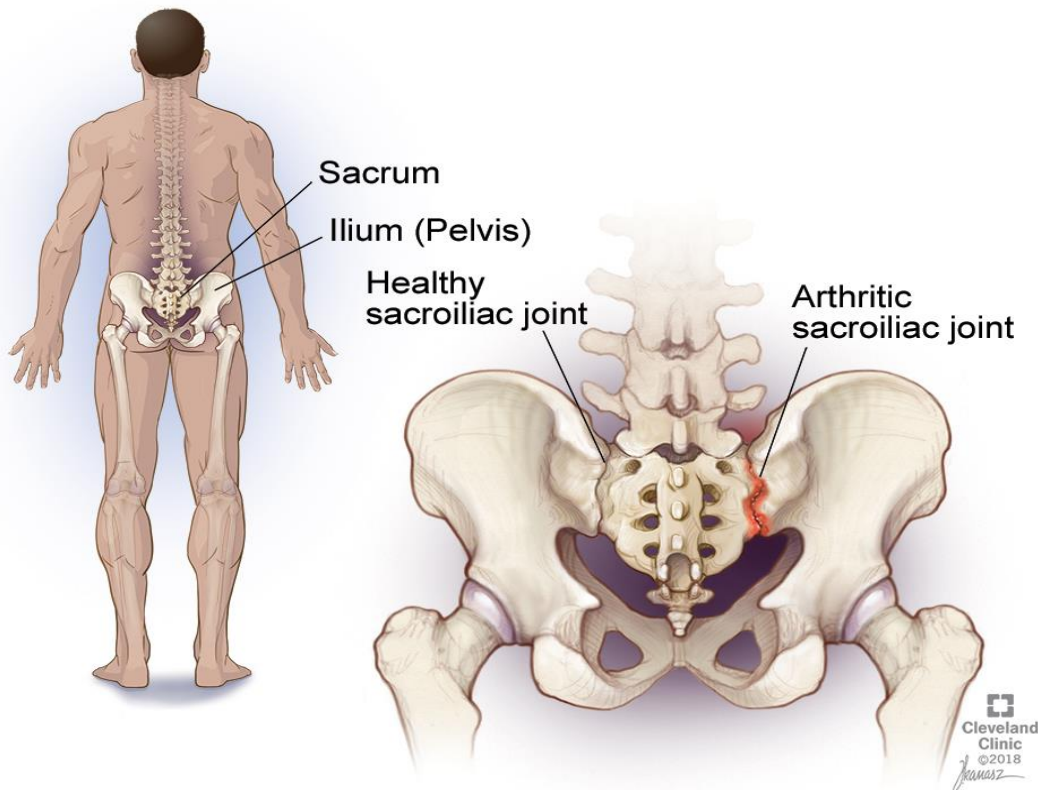
Psoas major muscle and piriformis muscle maintain the balance of tension of SI joint



Pathology of sacroiliac joints

Sacroiliac pain (lower back pain) & alteration of motion can be aggravated with prolonged **sitting** or **standing**, standing on one leg, stair climbing, going from sit to stand, and with running.

Potential causes of sacroiliac pain include **osteoarthritis**, traumatic injury, **pregnancy** and post-partum, systemic inflammatory conditions, and infection .



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Radiological examination of Vertebral column

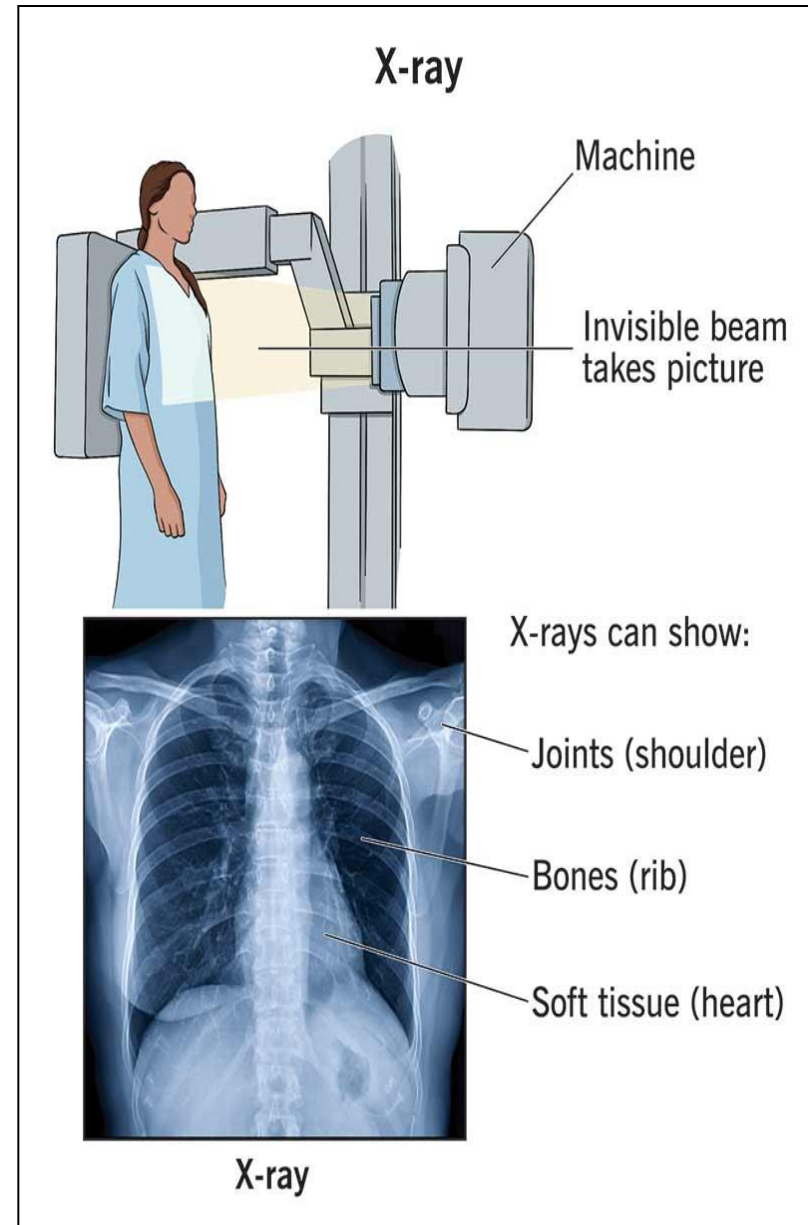
REMEMBER

What is Radiology?

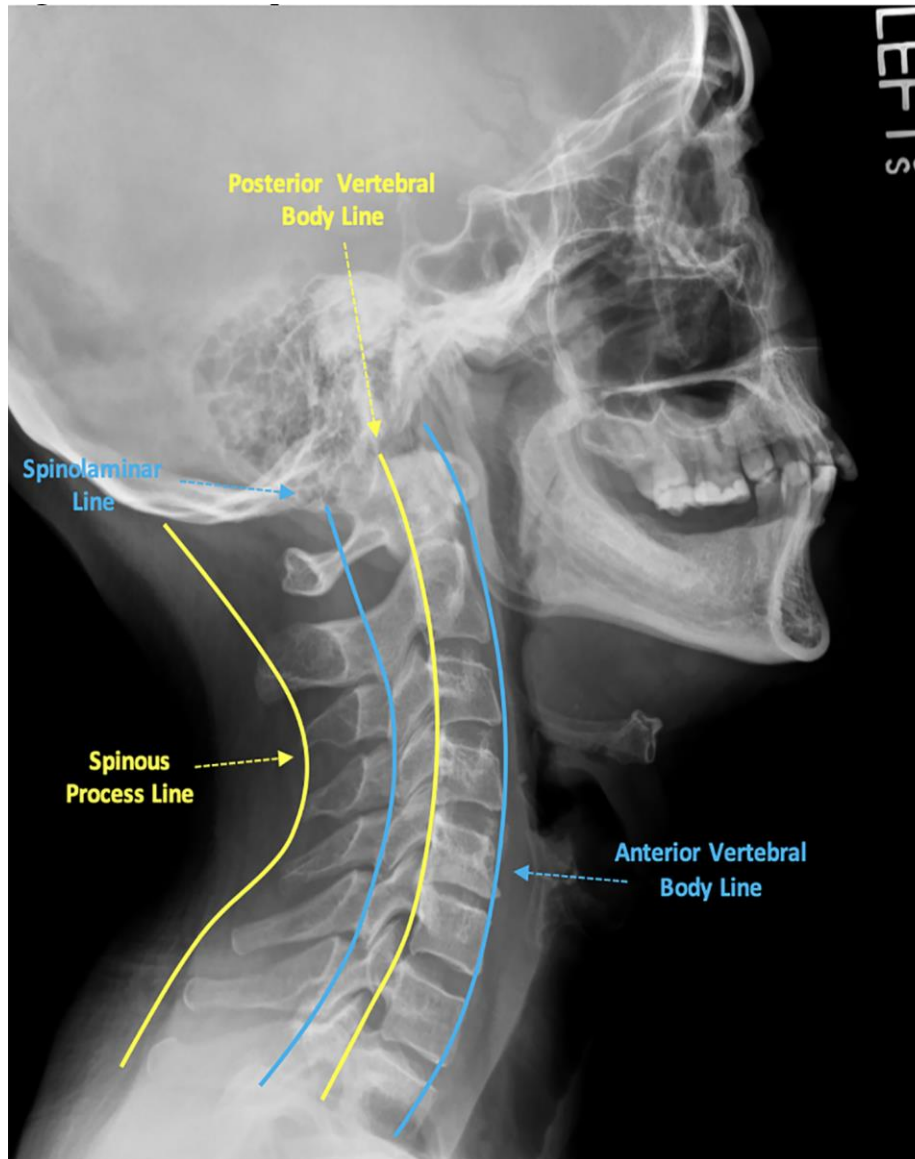
❑ It is a medical specialty that uses imaging to diagnose and treat diseases seen within the body.

Common imaging techniques of skeletal system

1. X-ray radiography : for bone fractures , degeneration or joint dislocations
2. computed Tomography (CT): for bone pathology . It offers detailed images, But **NOT** sensitive as **MRI** for certain bone conditions. .
3. Magnetic Resonance Imaging (MRI): for soft tissues pathology like tumors & metastasis . **Most sensitive and specific technique**

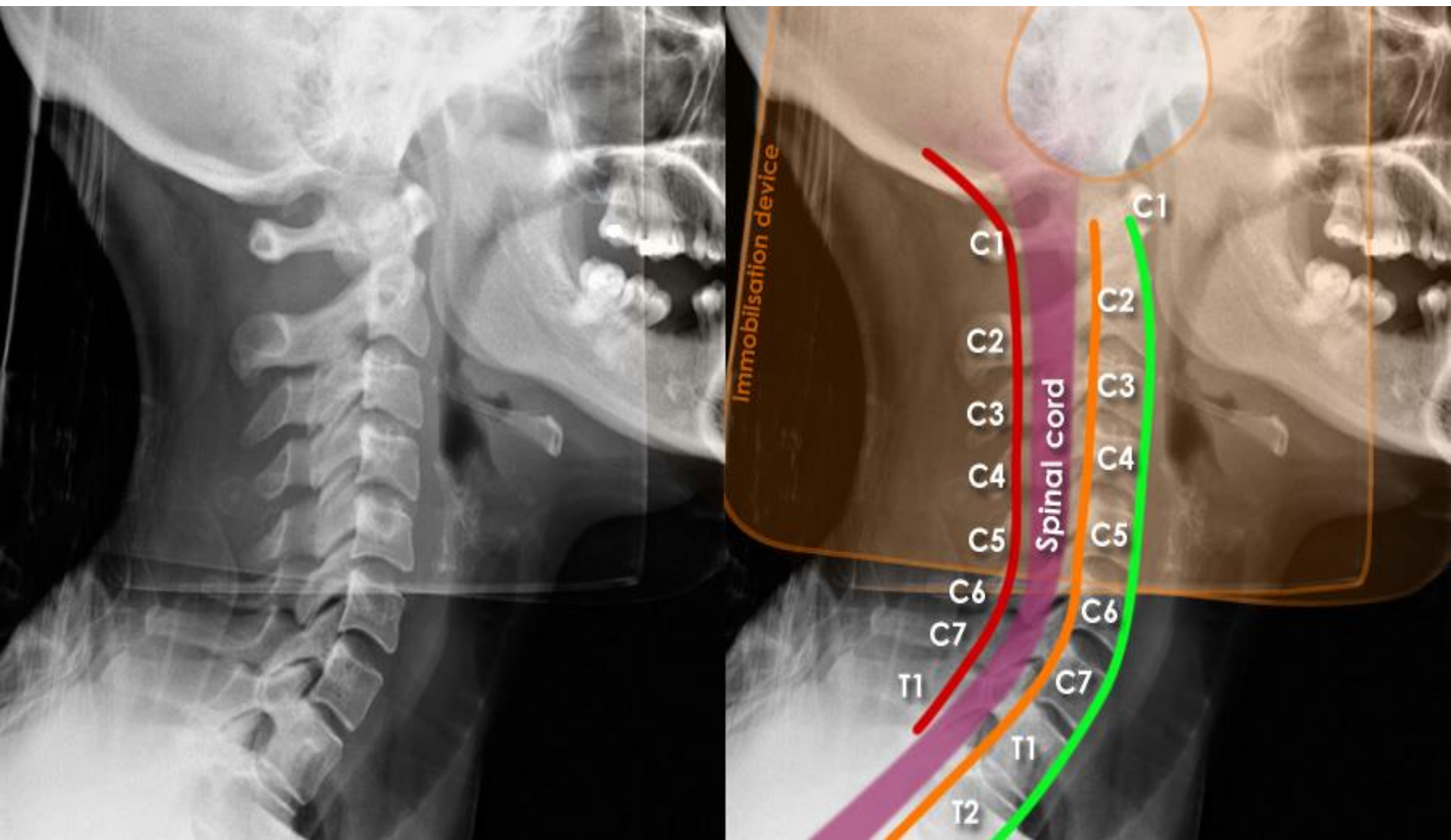


Plain X-Ray of skull and cervical vertebrae

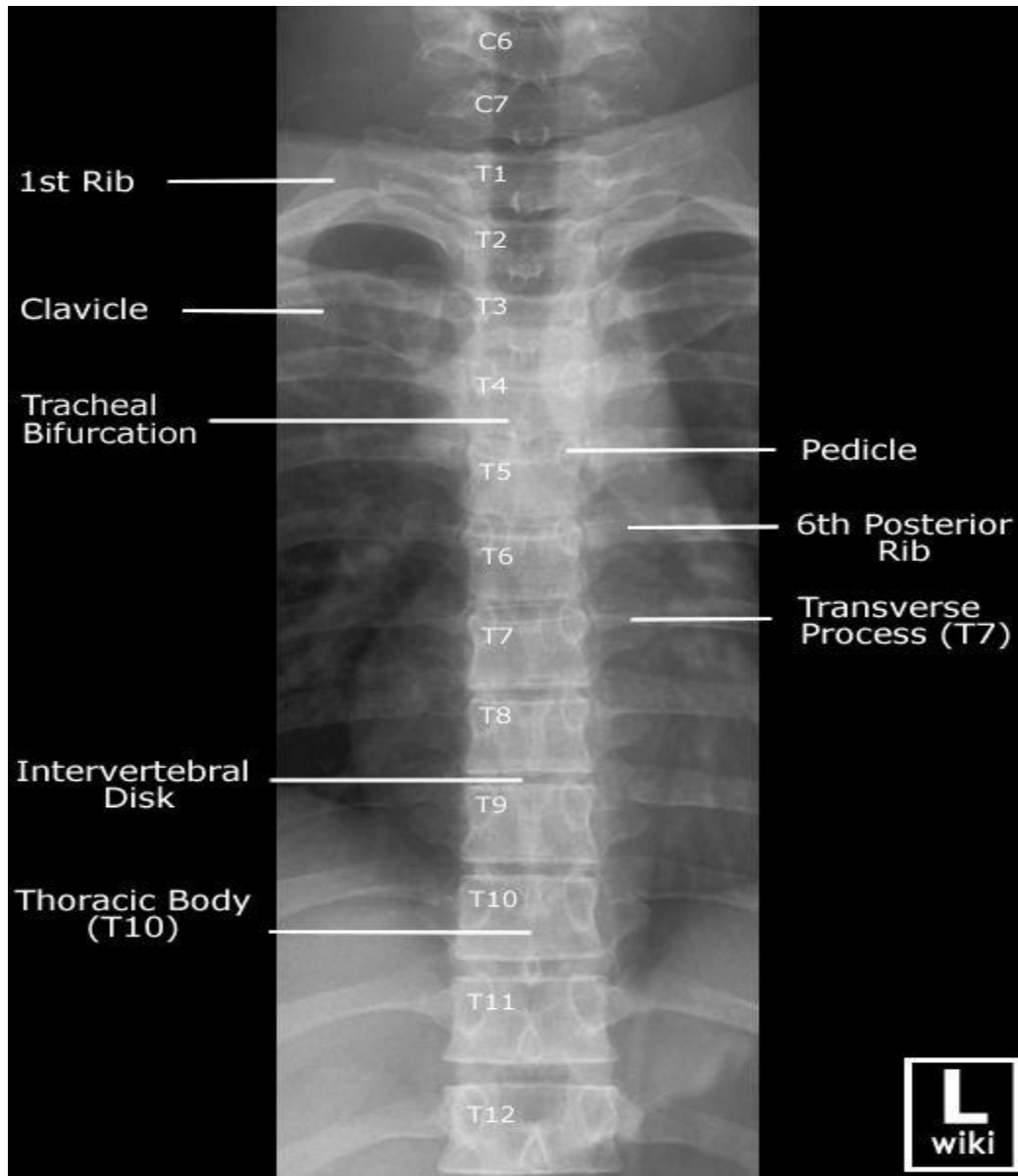


Lateral view

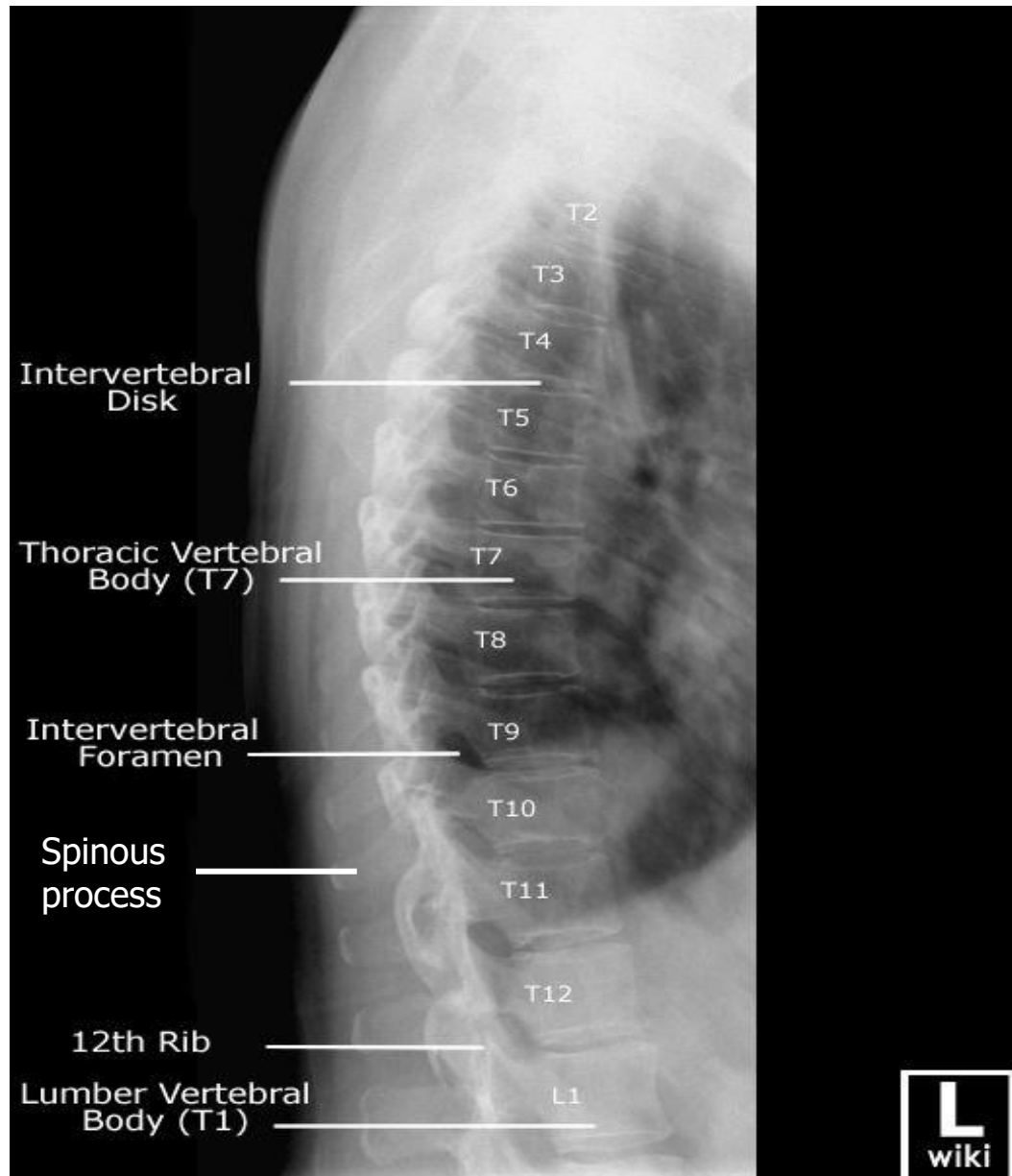
Lat view -of skull and cervical vertebrae



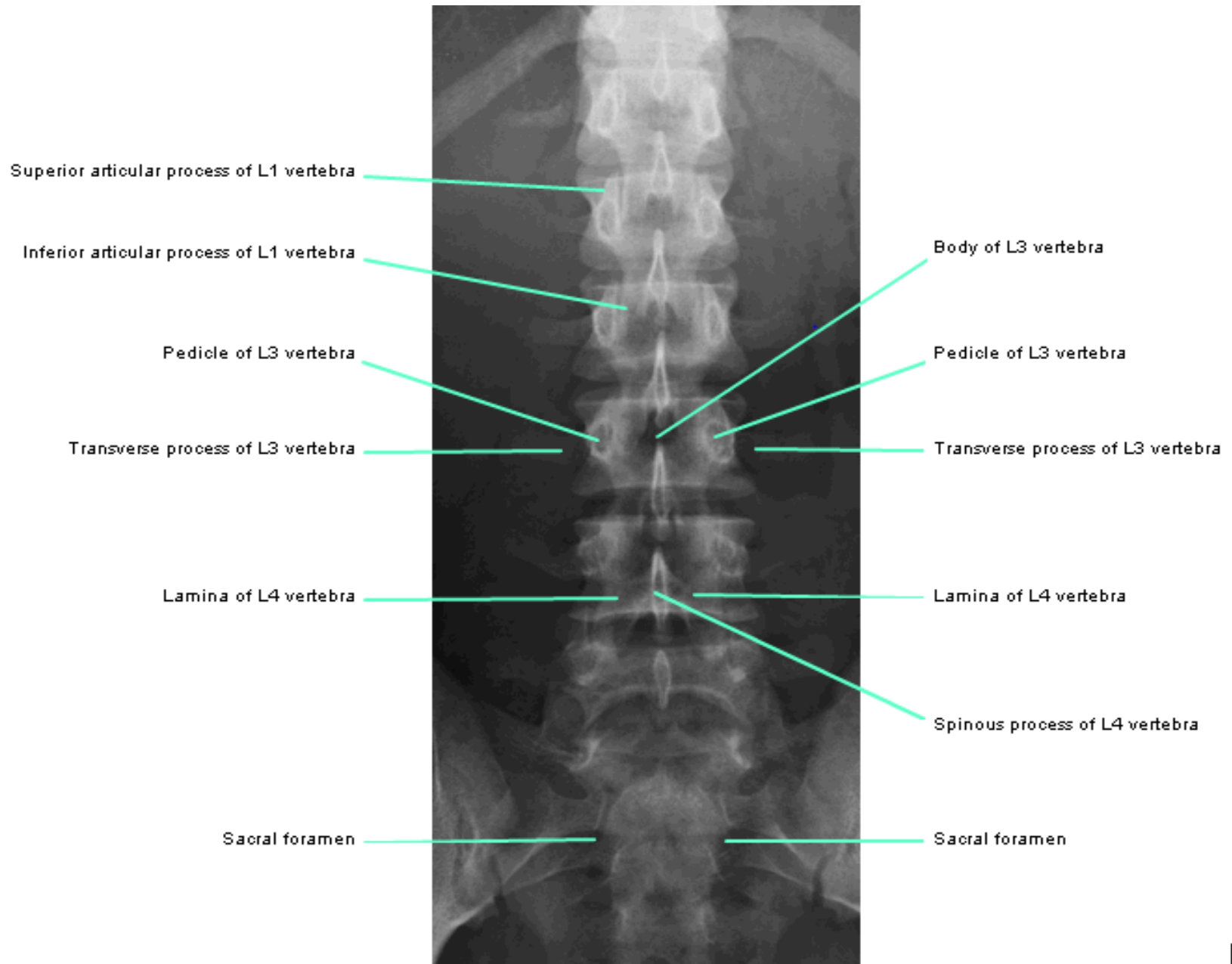
AP view of thoracic vertebrae

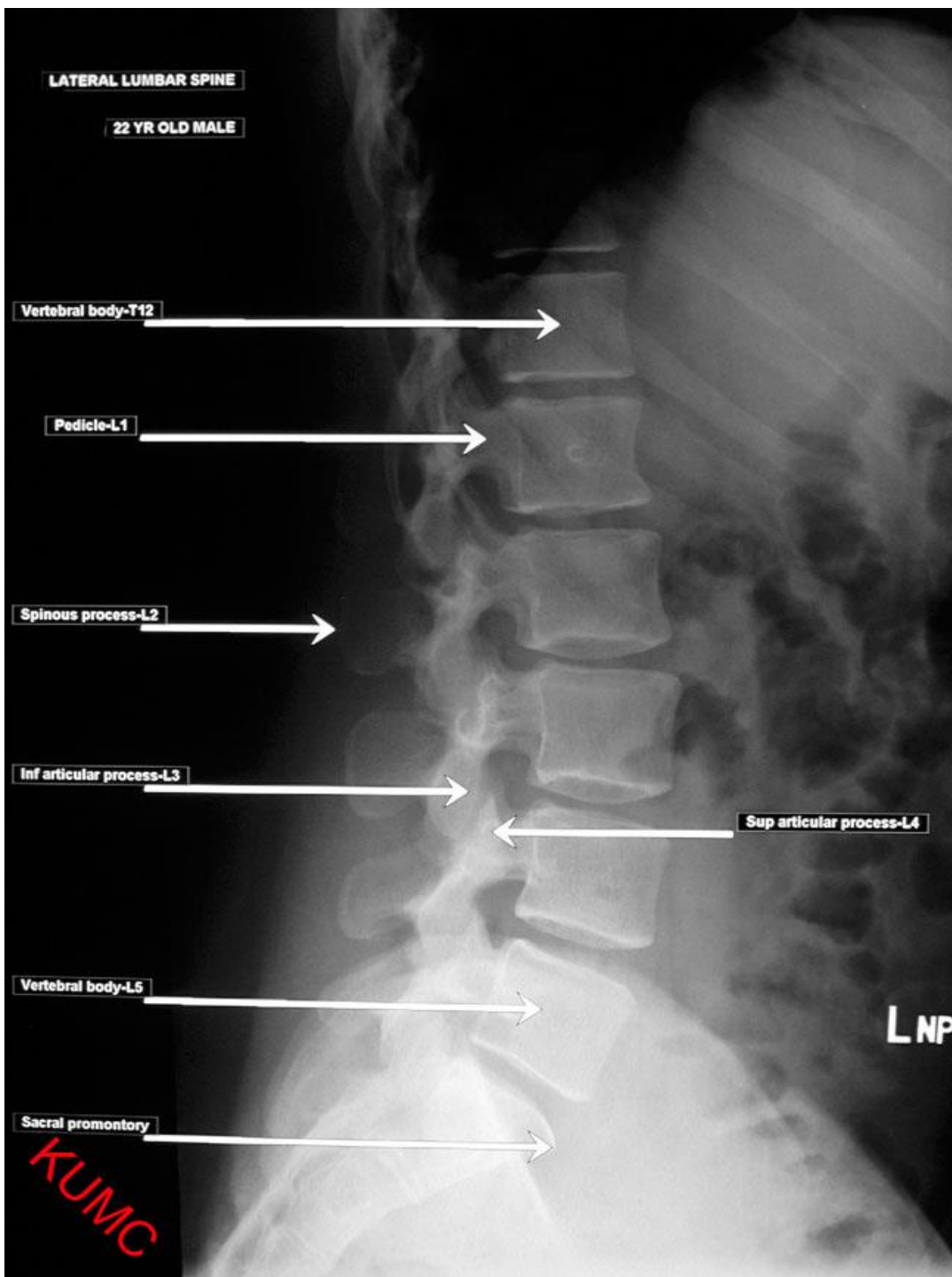


Lateral view of thoracic vertebrae

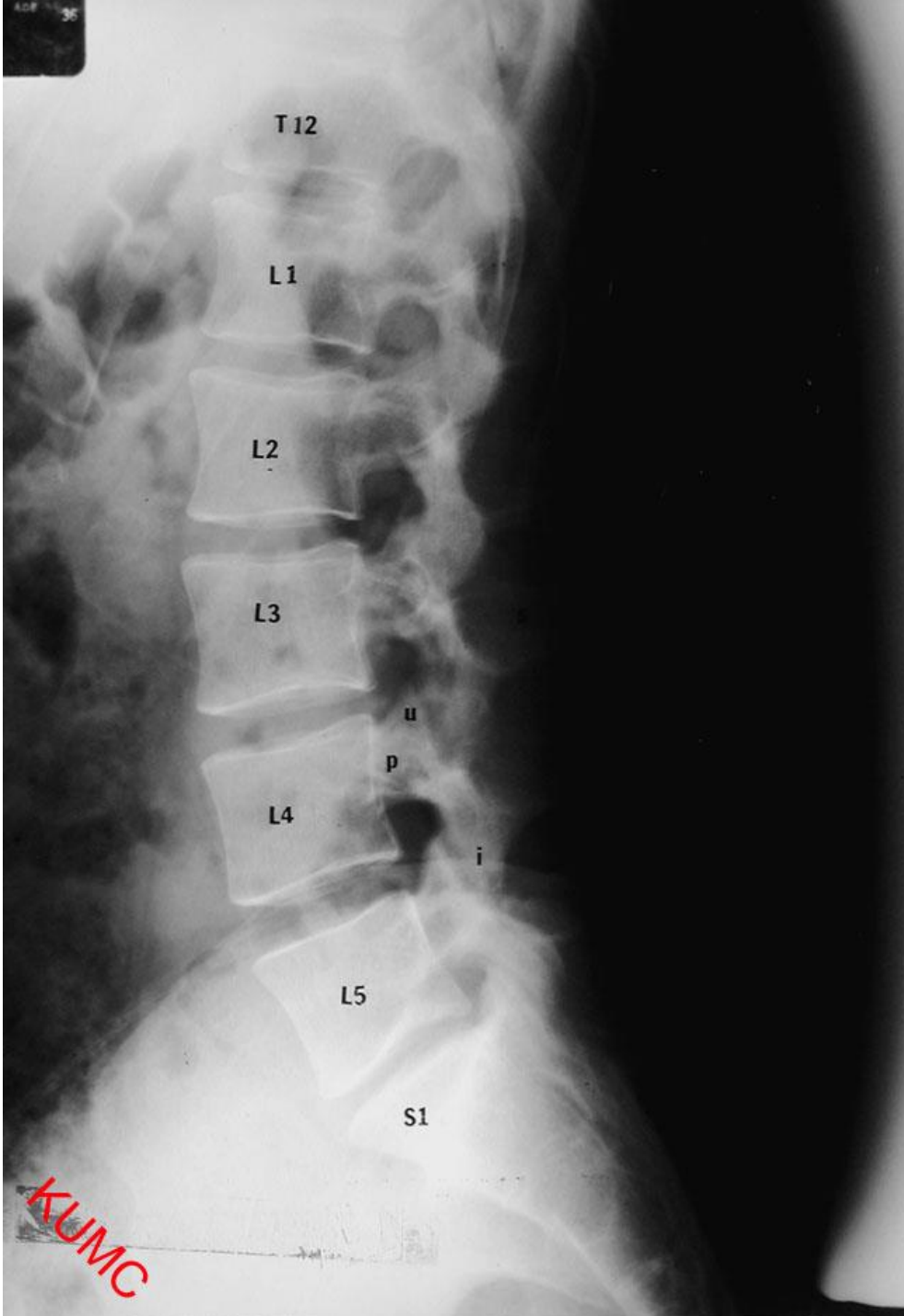


Lumbar Vertebrae: Anteroposterior Radiograph





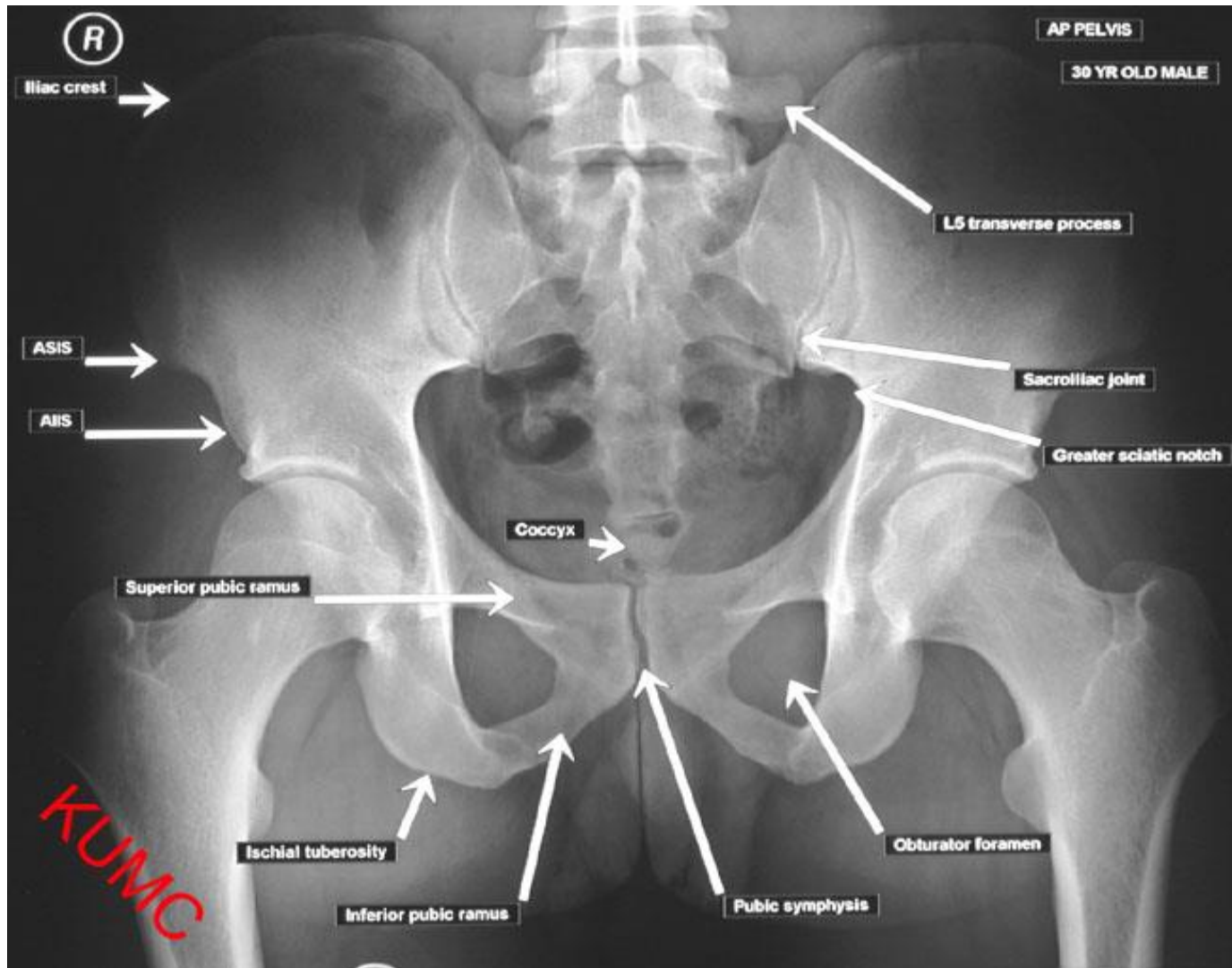
Lateral view of Lumbar vertebrae



Lateral view of Lumbar vertebrae

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AP view of Pelvis



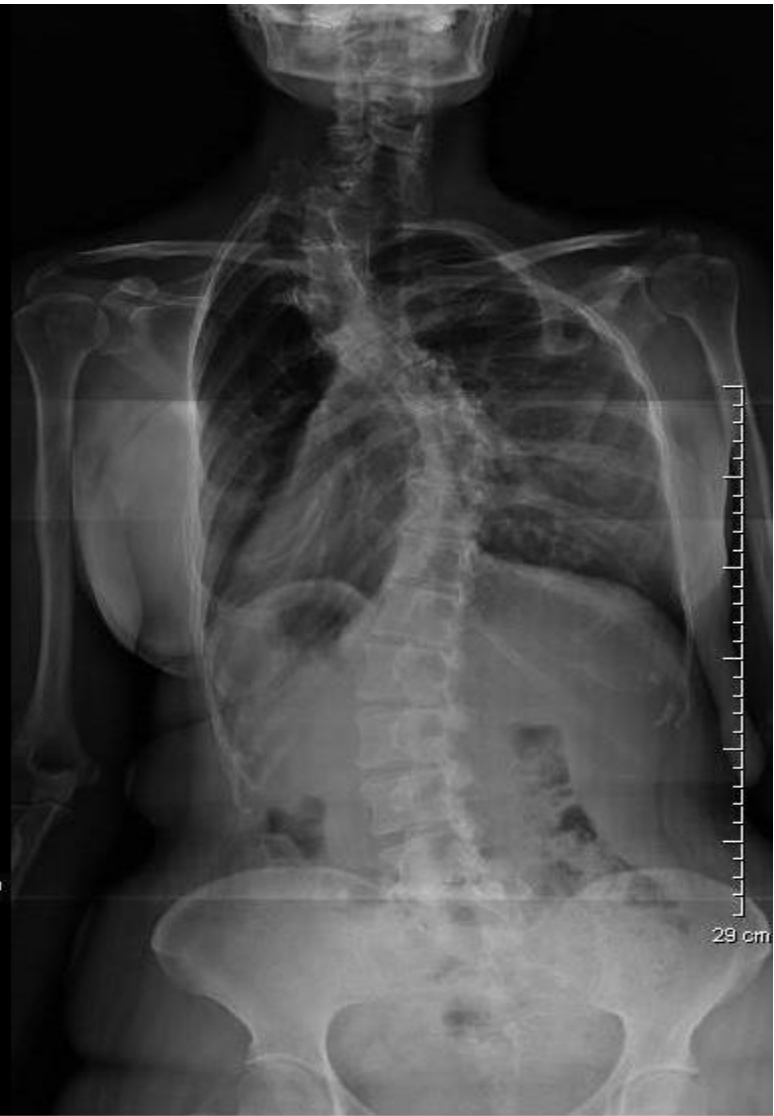
AP view of Pelvis



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Normal Spine Alignment



Double Thoracic Scoliosis

Radiography of congenital scoliosis



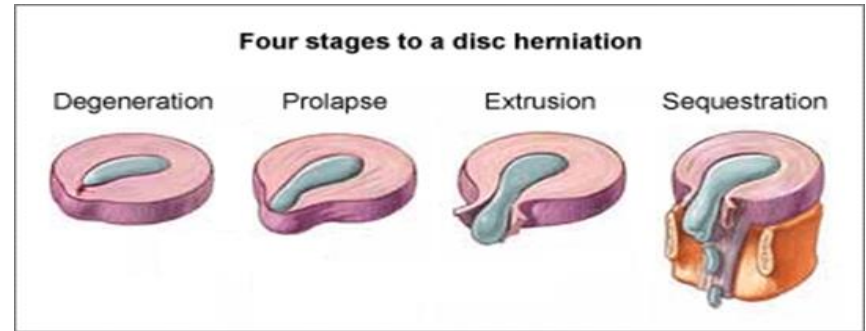
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Common causes of vertebral column injuries

(common in cervical & lumbar regions)

A. Intervertebral disc :

1. Degenerative disc disease leads to spinal stenosis
2. Disk bulge & herniation

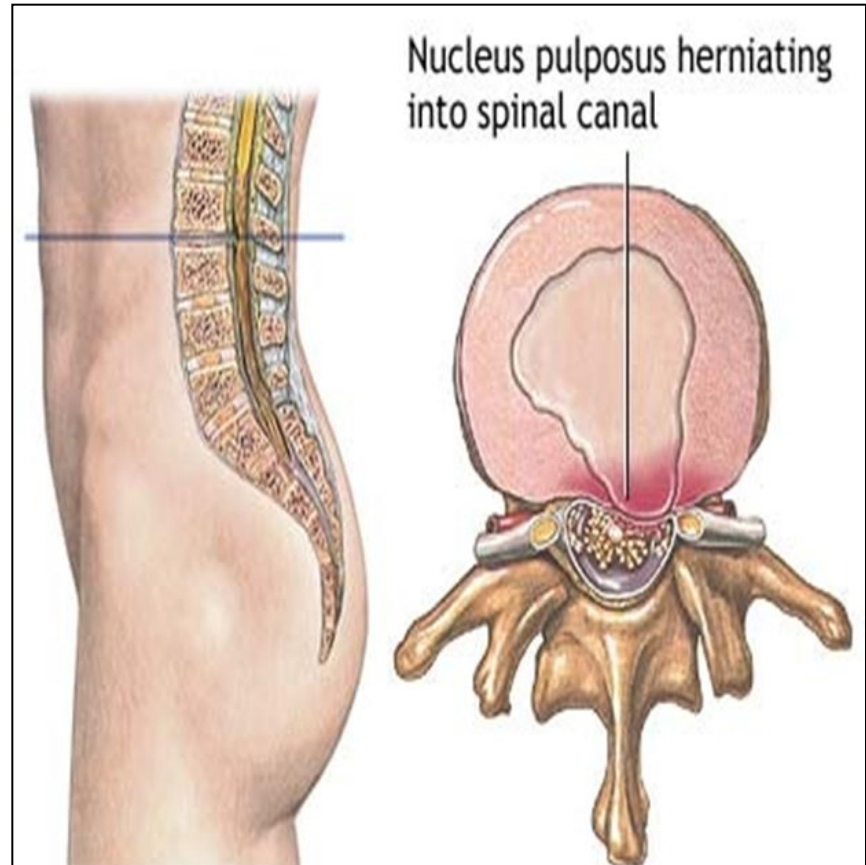


B. Joint : arthritic conditions:

1. Rheumatoid arthritis
2. Degenerative joint disease (Osteoarthritis) leads to spinal stenosis

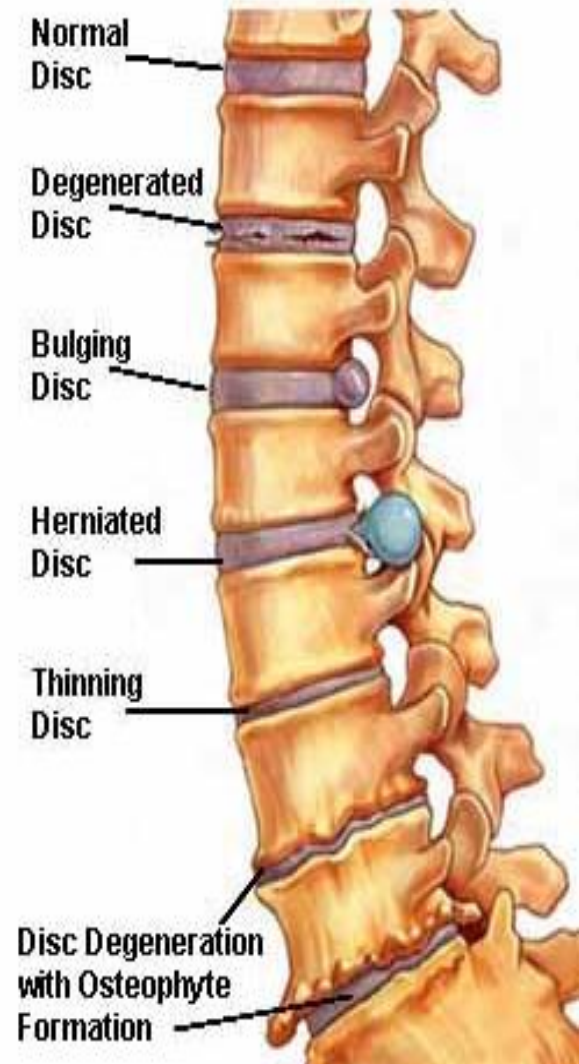
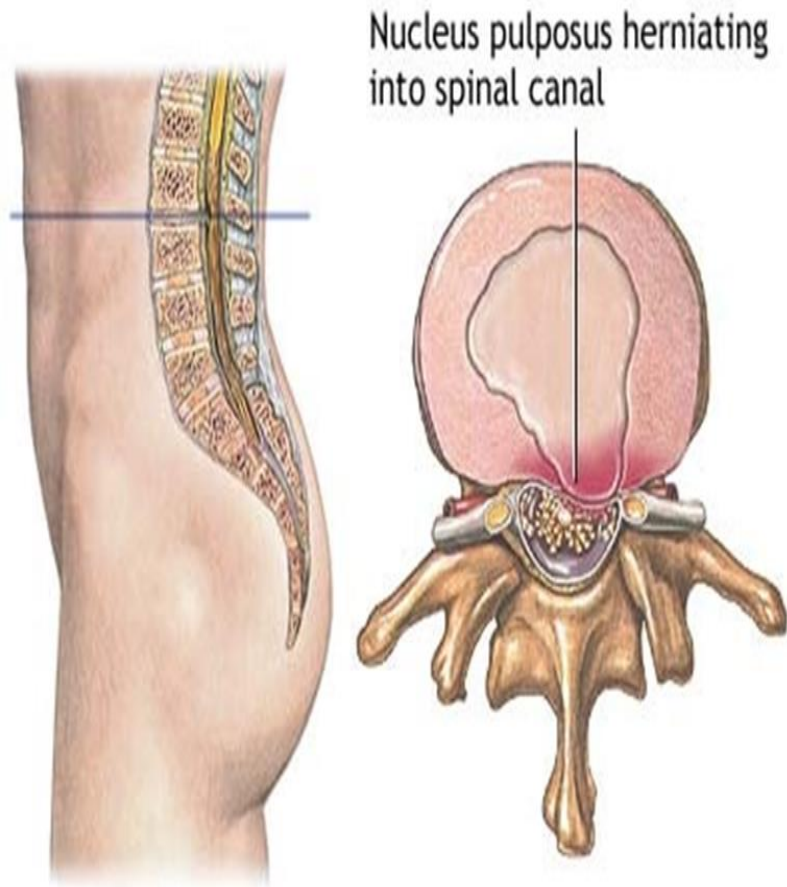
C. Bone :

1. Trauma causing fracture
2. Infection
3. Malignancy



Common injuries of Intervertebral disc :

1. Degenerative disc disease leads to spinal stenosis
2. Disk bulge & herniation



Disc diseases and spinal stenosis

Spinal stenosis

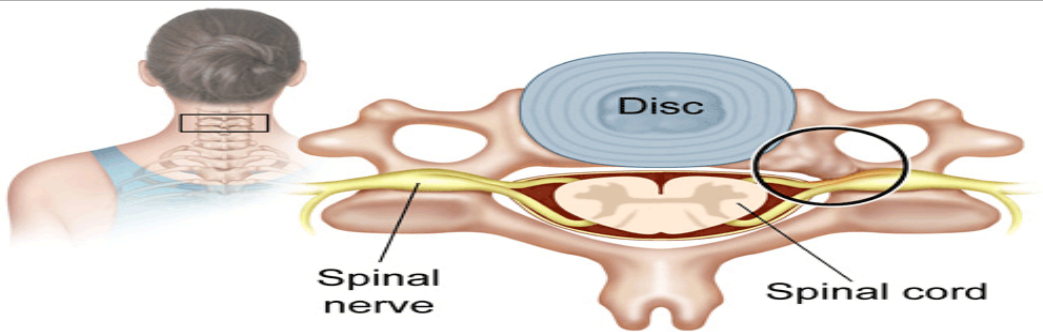
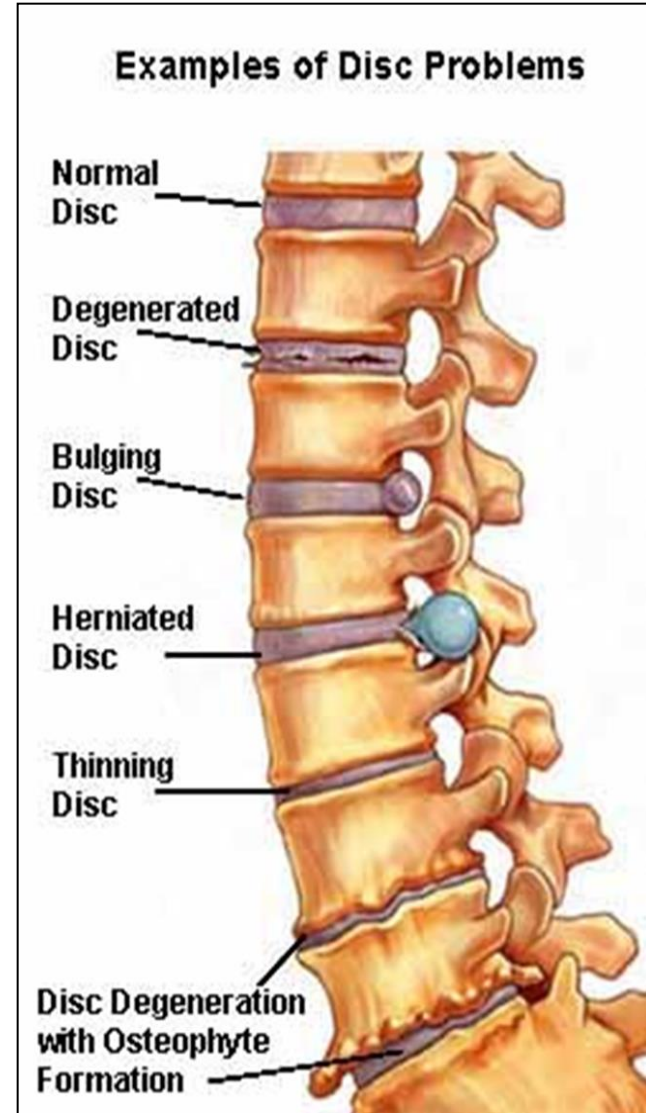
There are 2 types :

1. Central stenosis :

narrowing of the central canal resulting in pressure on the central part of spinal cord

2. Foraminal stenosis :

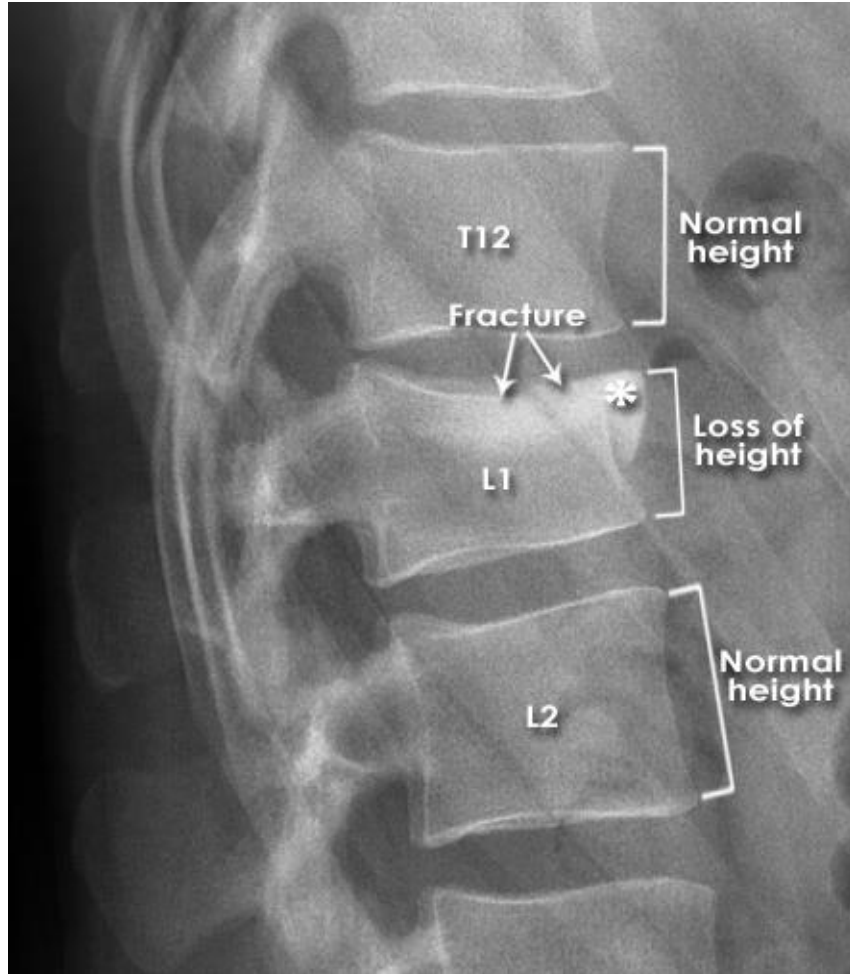
Narrowing of the intervertebral foramen resulting in pressure on the exiting nerve root



Disc herniation with spinal stenosis in lumbar region is the common cause of back pain & sciatica

Plain X-Ray radiographs

Fracture of thoraco-lumbar vertebrae
Lateral view



Fracture of lumbar vertebrae
AP view

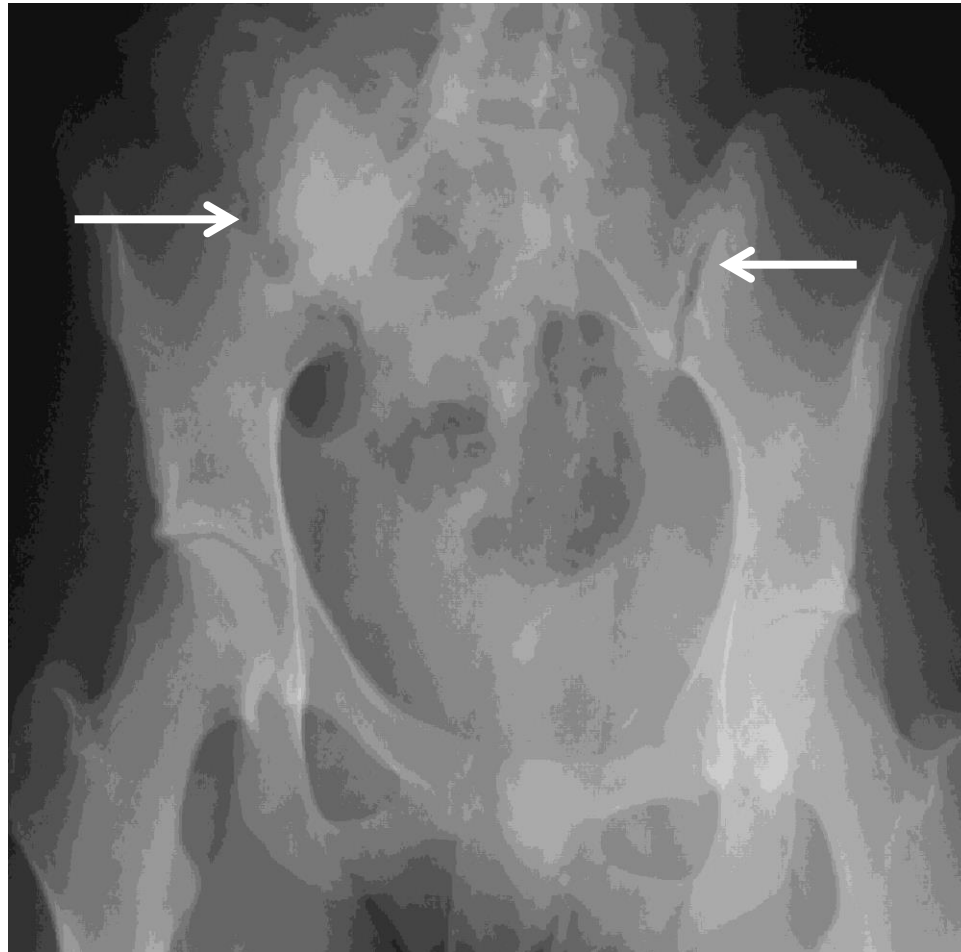


Plain X-Ray radiograph of Tumor of spine



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Plain X-Ray radiograph of Fracture & dislocation of sacro-iliac joint



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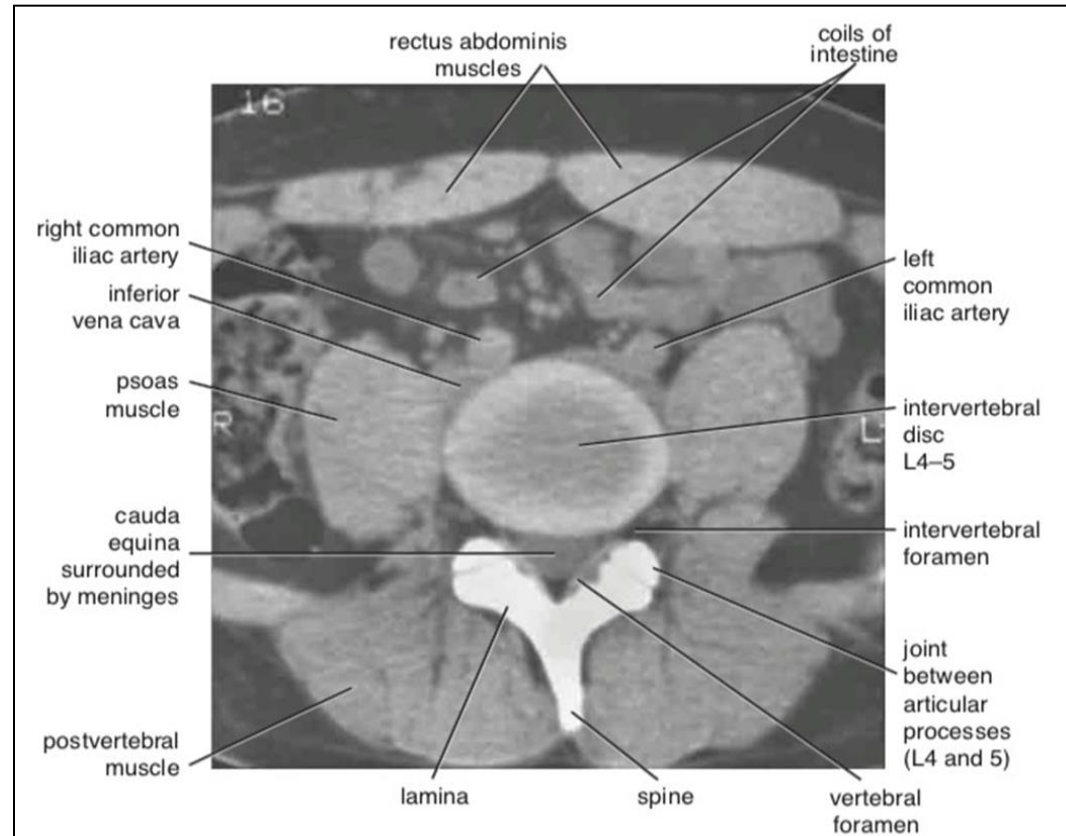
Computed Tomography (CT)

CT and MRI are extensively used to detect lesions of the vertebral column, **especially those involving the soft tissues pathology**.

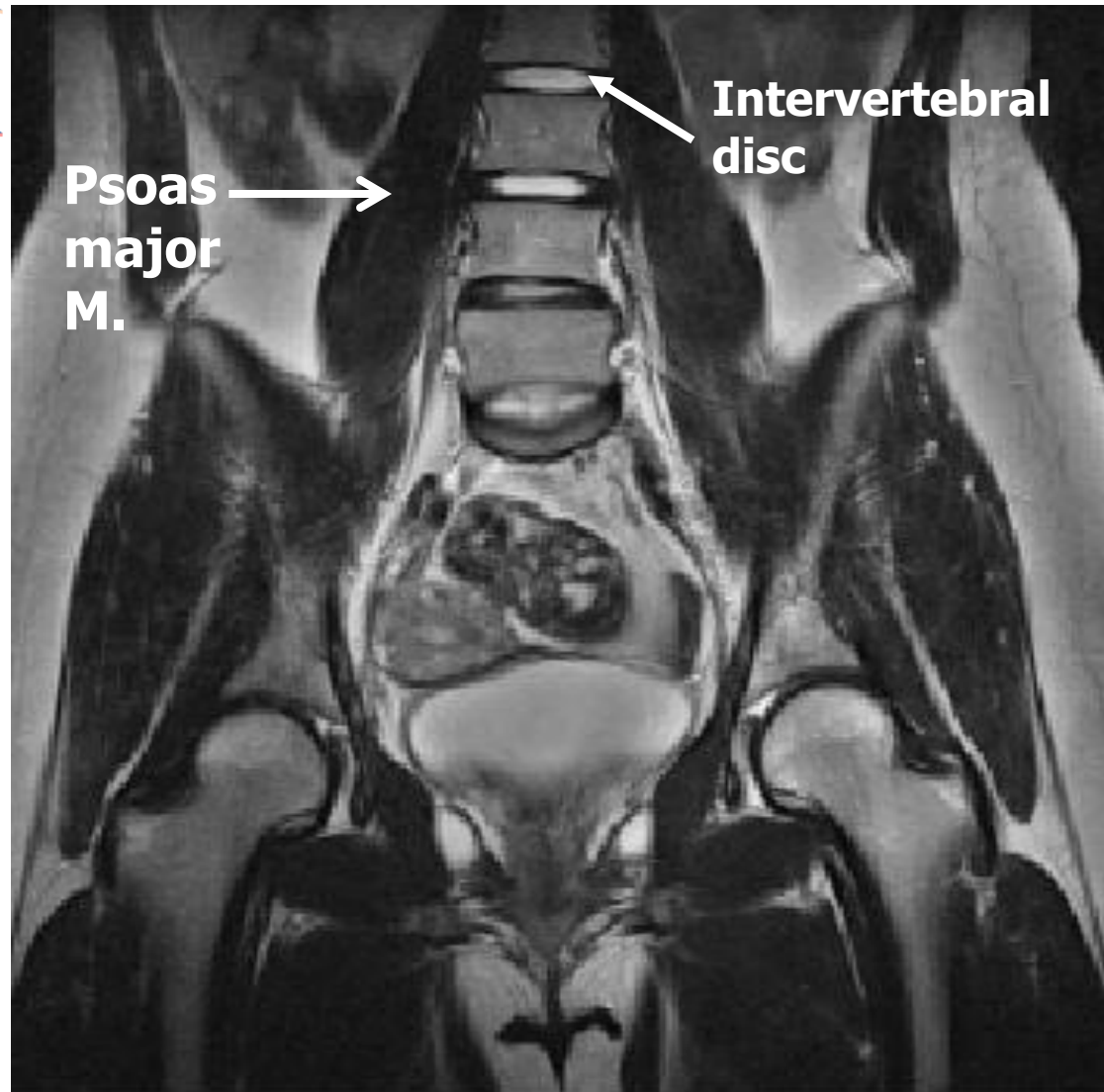
MRI is more diagnostic than CT scan.

CT scans can concentrate on the **intervertebral spaces** and reveal the **intervertebral disc** in transverse slices. The disc has a higher density than the cerebrospinal fluid in the subarachnoid space and the surrounding fat.

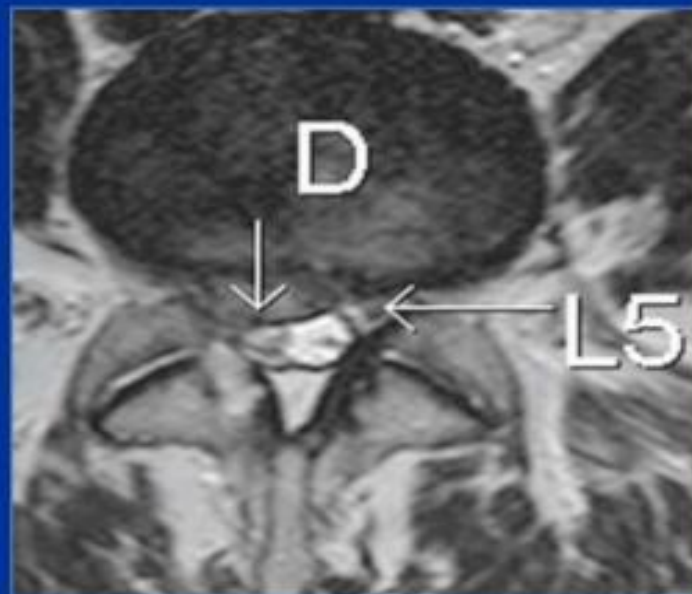
Fragments of a herniated disc can be identified beyond the boundaries of the annulus fibrosus.



Pelvic MRI

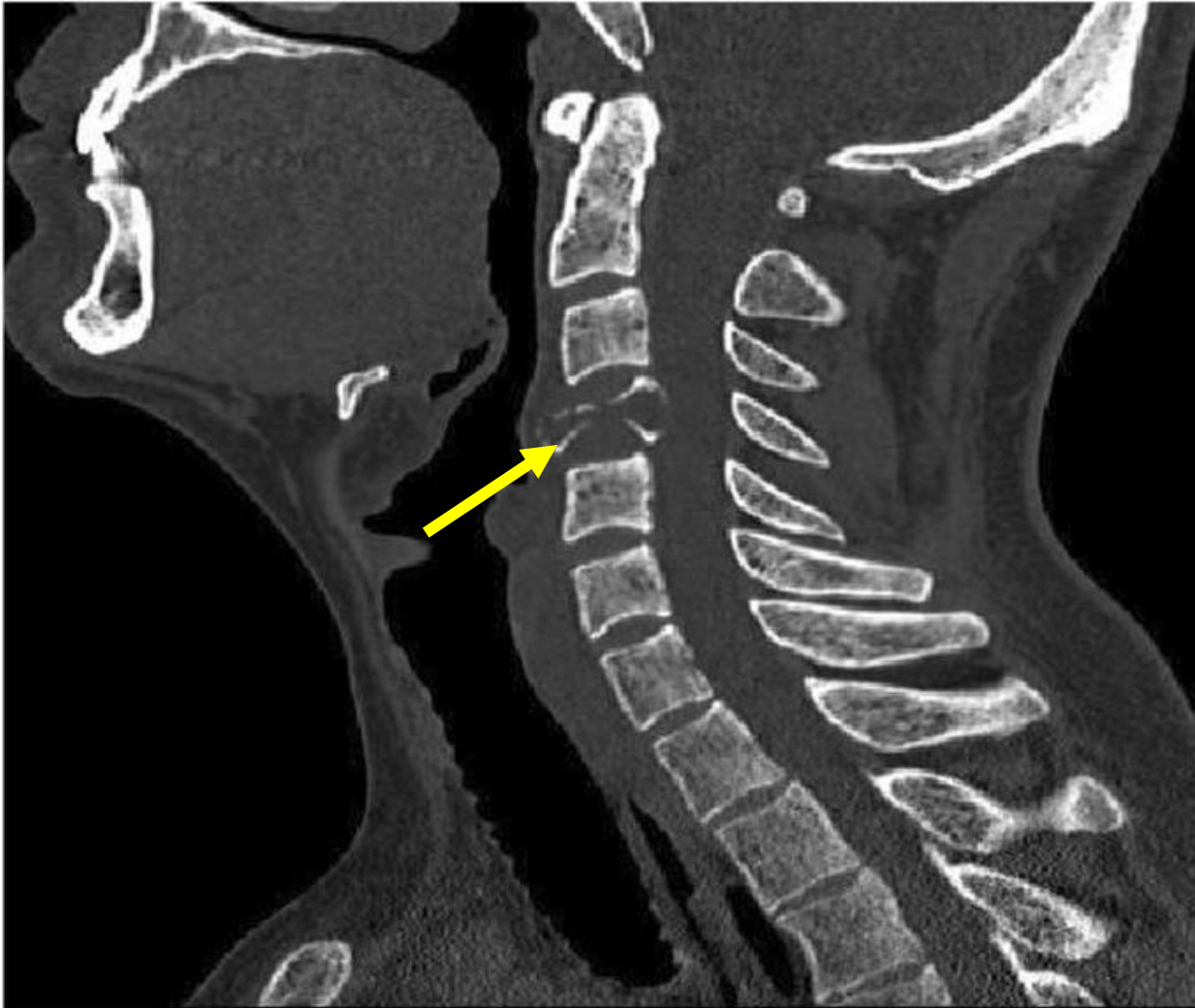


MRI is the choice in Inter-Vertebral Disc Prolapse (IVDP)



- MRI is the choice.
- Posterior herniation of the L4/5 disc.
- Transverse image:- Herniation into the right side of the spinal canal.
- Right L5 nerve root is compressed.

MRI – Malignant tumor on cervical vertebra



MRI –Spinal epidural abscess



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